VII primer v10.0

Vertically Integrated Industries: VC primer

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The idea in plain English. America has spent forty years outsourcing the “hard middle” of making things: digging and refining minerals, making metal, machining and welding parts, and producing chemicals. Now defense, industrial policy and China’s export controls make that middle valuable again, and it is short of plants and people. We look for startups that **own the plant and sell the physical output** (a mineral, a metal, a part, a chemical) and use **AI, software and automation inside their own operations** to build faster and run with fewer scarce skilled workers than incumbents. Owning production is what lets them keep the margin and the data.

The five industries (chapters 3–7). 1. **Raw Materials (Minerals & Metals)** — Stage A, finding and mining ore; Stage B, refining and making metal (smelting, alloys, magnets, powders). 2. **Manufacturing** — metal supply / service centers, then **fabrication (sheet/structures)** and **precision (machining, 3D printing, castings, composites)**. 3. **Specialty Chemicals** — producers of pharma ingredients and intermediates, agrochemical actives, electronic chemicals, fragrances, coatings and additives, catalysts and performance polymers, plus AI molecule discovery. 4. **Advanced Materials** — producers of engineered materials (battery cathodes, cooling fluids), with a separate **Materials Discovery Labs** sub-group. 5. **Enabling Software & Robotics** — tools sold to other people’s factories; tracked, **outside the core**.

Three labels, used for every company. - **Fully integrated** — owns production and sells the product itself. *Core fit.* (Example: SendCutSend owns its lasers and ships parts.) - **Hybrid** — owns some production or a lab, but also sells services, licenses its process, franchises, or relies on partners to make the product. *Partial fit.* (Example: Hadrian runs its own plants and also puts cells inside Lockheed’s plant.) - **Enabler** — sells software, design, robots or tools to other companies’ plants. *Outside the core;* tracked as competitor, supplier or comparable. (Example: Arda.)

How to read this doc. Every industry chapter has the same order: (1) the problem today, (2) macro and tailwinds ranked and marked structural or reversible, (3) headwinds at equal weight, (4) why an AI-enabled vertically integrated company is the answer and where that breaks, (5) what would have to be true and signals — then deeper macro detail, how the industry works, unit economics and data loop, company profiles grouped by label, bear case and key sources. The macro is deliberately the heaviest part. Statements are kept apart: - Plain sentence ending in `[src-…]` = sourced fact. - **Claim (who):** = a company, founder or expert assertion. Sourced, not verified. - `[inference]` = my read. Not sourced; argue with it.

Confidence (high/med/low) is marked on weak numbers. Acronyms are collected in Appendix A. Excluded by Lilly (failure examples only):

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0. Decision summary

In one line: the category is real and defense demand is strong, but most buyers do not pay much for speed alone, factories must be built before contracts arrive, and nobody has yet shown that the software makes each part cheaper or better over time.

Verdict [inference]. Back capital-light owner-operators at Series B only where speed and qualification are already paid for (Senra, Freeform, Isembard, as diligence hypotheses). Use Hadrian, Divergent and SendCutSend as benchmarks. Do not pay software multiples for job-shop economics until a company shows unit cost or quality improving with volume. Nobody in the repo shows that yet.

What the evidence says (102 podcasts and 28 expert calls, read in full). Each point is a speaker’s claim unless marked; detail sits in chapter 4. 1. **Price is not the wedge; speed is, and only for some buyers.** Hadrian’s CEO: “most of the market just doesn’t value speed. The market very much values flexibility, scale, total program cost” [src-20260921-876080]. Three Boeing voices say they pay parity or more (“we have actually paid probably more”) [src-20260818-9b2d5c][src-20250712-e5e110]; a Northrop GM: “Cost is secondary. Cost is driven usually by schedule” [src-

20250707-3b7a46]. The only “about 50% cheaper” report (Raytheon) was RFQ pricing before delivery, brokered by a Hadrian investor [src-20250708-2aa3ad]. 2. **Capex comes before contracts. Claim (Chris Power):** “0% chance of growing revenue unless you spend a billion in CapEx” [src-20260921-876080]; also “close to a billion in revenue next year”, about 30% “pure software” (a stated plan, not a reported number [inference]) [src-20260610-779f8b]. Debt for machines, equity for software (section 2.5). 3. **Cheap labor is not the edge.** Labor is “less than 10% of COGS” in a new automated plant; China wins on a co-located supply chain [src-20260513-bdc044]. SendCutSend can beat China on only about 20-25% of its work [src-20260519-2d6ceb]. High utilization comes from scheduling software, not robots (SendCutSend lasers at 86%) [src-20260626-541ac8]. 4. **Quality and qualification are the gate.** “None of their parts right now are flying” (Boeing Defense) [src-20260818-9b2d5c]. OTA prototype contracts waive many mil-spec rules; in production “there’s a gap” in lot inspection Hadrian “is still maturing” on [src-20260908-ed94f0]. The Navy: “we have an adoption problem” [src-20250319-f21bec]. Entry is through repair and replacement parts [src-20251015-e9edbf][src-20260821-1c5a08]. 5. **Share per customer is capped and the edge leaks.** SpaceX caps a shop at about 1/3 of its revenue [src-20250708-342eaf]; Boeing “will never” put one part family on one supplier [src-20260818-9b2d5c]. SendCutSend customers “outgrow us” [src-20251002-6a5afd]; enablers (GrayMatter, CloudNC) sell the same automation to incumbents [src-20261022-3335bd][src-20261009-7cafd0]. 6. **The only learning-over-time number** is Hadrian’s: “within a couple of years, we saw a 50% cut” in order-to-first-article time; its software “is replicable” and the moat is “financial power” plus a head start [src-20261004-9b47f5]. 7. **Benchmarks and demand.** Incumbent roll-ups run “effectively … a 30% gross margin” and buy at “seven times EBITDA” [src-20241001-28c2aa]. The FY26 NDAA adds multiyear munitions buys and advance-payment purchase commitments (Secs. 804(b), 867), the first channels that could turn prototype wins into multi-year orders [src-20251218-f0d6ef] [inference]. Materials R&D budgets have not grown “crazily” [src-20260909-af3e60] (low).

Evidence on the four thesis claims (claims as numbered in insights/thesis.md)

#	Thesis claim	For	Against	Read [inference]
1	Demand for domestic production is structural	Lockheed 7-year cell deal [src-20261014-015327]; Navy $900M [src-20260320-67ba4c]; Thales CPO: capacity is the pain after COVID-era supplier bankruptcies [src-20251110-84b67e]; Boeing has used parts taken from non-flying aircraft [src-20260818-9b2d5c]	34% spindle utilization at existing shops [src-20250120-419768]; Taiwan 20-30% cheaper and Vietnam/Malaysia cheaper still [src-20250709-43486e]; Boeing makes about 40% of parts in-house and holds 6 months of inventory [src-20250712-e5e110]	True for defense capacity and speed. Commercial rate demand not shown
2	Owning the plant creates a data loop	Freeform and Mariana score 2 (section each chapter’s data-loop section). Ex-Hadrian: about 80% of non-CAM labor automated, lead times cut about 50% [src-20250708-342eaf]; reprogramming saved 50-80% of CAM time [src-20260611-7b80f0]	Toolpaths come from third-party CAM, with Vericut (simulation software) as the check [src-20260611-7b80f0] [src-20261004-9b47f5]; CMM (coordinate measuring machine) inspection is manual [src-20260817-8867f2]; a buyer asks “is it actually” software-enabled [src-20250729-a0a78c]	Workflow automation is real. Learning with volume is not evidenced

#	Thesis claim	For	Against	Read [inference]
3	Moats: process data, secured capacity, qualification	Trusting a new supplier takes 5-10 years [src-20250712-e5e110]; change-of-control clauses in defense contracts (contracts that can be voided if the supplier is sold) [src-20260817-8867f2]; lock-in from 50-year programs [src-20260818-9b2d5c]	Share caps (finding 3); 90% of the first 10 Hadrian employees gone by year 5 [src-20260817-8867f2]; incumbent PCC is also automating [src-20250712-e5e110]	Qualification is a moat once paid, but it caps share. No data moat evidenced
4	Hardware-inclusive producers can return venture outcomes	SendCutSend profitable 8 years [src-20260519-38c7ed]; Hadrian plants are debt-funded [src-20260817-c48f28]	“Not … a fantastically profitable business”; a six-machine plant 1 cost “well north of $10 million” [src-20260817-8867f2]; net per-part saving after capex is 20-30% at best [src-20250712-e5e110]; Hadrian is “better suited to debt than VC” (ex-Anduril PI) [src-20250709-43486e]	Unproven. Underwrite the speed premium and the debt capacity, not cost leadership

0.1 Views at a glance *(analyst judgment, not fact [inference]; detail in each chapter’s “Our view” box)*

Category	Stance	Conviction	Top names (label)	Key signal
Raw Materials	Selective	Medium	Vulcan, Mariana (Full)	Offtakes renew without subsidy
— Stage A, exploration	Watch	Medium	Earth AI (Hybrid)	First formal resource estimate
— Stage B, refining and metal	Attractive, selectively	Medium	Vulcan, Mariana (Full)	Mariana hits nameplate; Section 232 holds
Manufacturing	Selective	Medium	Senra, Freeform (Full); Isembard (Hybrid)	Unit cost falling with volume
— Metal Supply	Watch	Low	Nox (Full, passed)	AI distributor beats Reliance margins
— Fabrication	Attractive	Medium	Senra, 1872 (Full); Machina (Hybrid)	Customers stay as they grow
— Precision & Defense Hardware	Selective	Medium	Freeform (Full); Isembard (Hybrid)	Parts flying on a production program
Specialty Chemicals	Selective	Medium	Solugen, Osmo, Phlow (Full); Scimplify (Hybrid)	Phlow FDA approval in 2027
— Pharma ingredients / Fragrances	Attractive	Medium	Phlow, ODP / Osmo (Full)	FDA approval; large fragrance-house customers
— Agro / electronic / catalysts	Watch	Low	Enko (Hybrid); Dunia (Enabler)	A registered product or supply contract
Advanced Materials	Selective	Low–medium	Orbital, Mitra (Full)	Orbital fluid ships with 2027 GPUs

Category	Stance	Conviction	Top names (label)	Key signal
— Producers	Selective	Medium	Orbital, Mitra (Full)	Qualification and offtakes
— Materials Discovery Labs	Watch	Medium	Radical (Hybrid); CuspAI, Periodic (Enabler)	A lab sells material
Enabling Software & Robotics	Watch	Medium	Arda, Mind, GrayMatter (Enabler)	Retention; robot success above ~90%

1. The theme in plain English

What it is. AI-enabled vertically integrated industries means companies that build and run their own factories (or, in one variant, the software that runs other people's factories) and sell the output: parts, metal, material. The product is the physical thing; the software is how they make it faster and with fewer skilled people than incumbents. SPC's slide 8 frames it as producers "that build and run their own production capacity, using in-house AI and software to quote, schedule, and control robots and machines" [src-20261022-80fd75].

Legacy vs AI-native.

	Legacy	AI-native
Who	Job shops (small owner-run machine or fab shops that build to a customer's drawing), service centers (metal distributors that stock, cut and ship metal), brokers and marketplaces	Hadrian, SendCutSend, Senra, 1872, Nox, Mariana
Software	ERP and CAM bought off the shelf (CAM = software that turns a 3D model into machine toolpaths; ERP = the back-office system of record)	Built in-house around quoting, nesting (packing parts onto a sheet or block to minimize scrap), CAM automation, scheduling, robot control and inspection
Speed	Average aerospace part 8-12 weeks, 40% late or with quality issues [src-20251024-2e5383] (med, single memo)	**Claim (SendCutSend):** 48 hours design to delivery [src-20260227-e92d45]; **Claim (Senra):** harness lead time six months to two weeks [src-20260128-a16c79]
Learning	Knowledge sits in the heads of machinists averaging early-60s in age (**Claim, Chris Power**) [src-20260908-65fab6]	Thesis: every part made improves the next quote, schedule and QC. Evidenced at 2 of ~15 names (section each chapter's data-loop section)

Three business models (taxonomy `model`):

Model	Best example	Tradeoff
owner-operator	SendCutSend: owns all its lasers and plants, bootstrapped to a $1.01B round in May 2026 [src-20260519-38c7ed], then a Series B at $2.1B five months later [src-20261020-487a39] (med, single trade write-up)	Owns the data and the margin; carries the capex and utilization risk
software-first	Arda: CAD-to-work-instructions agent inside Anduril's ArsenalOS; $160M Series A at $1.5B [src-20261020-914a64]	Capital-light, fast to revenue; needs a host plant for data and competes with Siemens and Rockwell on the installed base [src-20250326-cc1810][src-20260804-0e8130]

Model	Best example	Tradeoff
hybrid	Hadrian (owns factories, now embeds cells inside Lockheed's Grand Prairie plant on a 7-year deal) [src-20261014-015327]; Isembard (owned plus franchised factories on MasonOS) [src-20260909-b547c7]	Tries to get owner data with partner capital; harder to underwrite because economics are split

The core tradeoff in one line: owner-operators buy data ownership and pricing power with capital intensity; software-first buys speed and margin with data dependence on someone else's plant [inference].

2. Cross-cutting macro (applies to all five industries)

The short version [inference]. Four things are true at once: 1. Washington is spending, lending and even taking equity at a scale not seen since the Cold War, for anyone who can produce metals, magnets, munitions and chips in the US. 2. Building a factory has got more expensive and more uncertain: rates went back up, metal tariffs raise input costs, and factory construction spending is falling. 3. China has shown it will cut off critical minerals, and the current pause on those controls expires in weeks. 4. Labor is the binding constraint, not demand.

AI-native producers sell into points 1 and 3 and are hurt by point 2. Their labor pitch (fewer, more productive people) only works if the software is real.

2.1 Manufacturing in one page

- **How big it is.** US manufacturing added $2.90 trillion of value in 2025, 10.1% of GDP, employing 12.65 million people at average pay of $112,261 [src-20261013-a7c995].
- **How it is organized.** A pyramid. At the top are a few **primes** that sell finished systems to the government: in FY2023 Lockheed Martin ($61.4B of contracts), RTX ($24.1B), General Dynamics ($22.9B), Boeing ($20.1B) and Northrop Grumman ($16.3B) [src-20251215-98f650]. Below are tier-1 suppliers (sub-systems), then tier-2 and tier-3 (parts, castings, forgings, harnesses, raw metal). Primes fell from 51 to 5 since the early 1990s [src-20220215-45c353].
- **The bottom is small and old.** 17,156 US machine shops; 83% have fewer than 20 employees [src-20250626-23f6d9]. 4,874 were lost from 2002 to 2018, about 20% [src-20250909-29c826]. **Claim (Isembard):** the average owner is over 65 and 40% plan to retire within five years [src-20260310-45bab3].

What this means [inference]. A fragmented, aging supply base under a few powerful buyers. That gap is the opening every company here is trying to fill.

2.2 Six forces behind "why now" (overview)

1. **Defense needs more munitions and ships, fast.** Concern about the industrial base "has focused on munitions production due in large part to high rates of expenditure" [src-20251215-98f650]. Detail: 2.6.
2. **China controls key inputs.** It leads refining for 19 of 20 strategic energy minerals, at about 70% average share [src-20250521-c62d29], makes about 94% of rare-earth magnets and cut magnet exports in April 2025 [src-20251103-ee9b27]. The US imports 100% of 16 mineral commodities [src-20260206-04fe3e]; almost 90% of the most advanced chips are made in Taiwan [src-20230928-c47558]. "Can we make this at home?" is now a security question, so governments pay for domestic capacity even when it costs more [inference]. Detail: 3.1.

3. **Tariffs make imported metal dearer.** Since April 2026, products made almost entirely of steel, aluminum or copper face 50%, products containing them 25%, and 10% if the metal was melted in the US [src-20260402-1b5d6c]. Nucor's margin rose from 14% to 20%, but its own fabrication arm reported "margin compression resulting from higher steel input costs" [src-20260812-77d6f5]. Tariffs are proclamations and can be reversed: a tailwind, not a foundation [inference].
4. **There are not enough skilled workers.** Detail: 2.7.
5. **The government now lends and buys, not just regulates.** Detail: 2.3-2.4.
6. **Money is available, but not cheap.** Hadrian raised $1.37B in one round [src-20260807-b0dc5b], but the Fed raised rates to 3.75-4.00% on September 16, 2026, saying "inflation remains elevated" [src-20260916-f0c016]. Detail: 2.5, 2.9.

2.3 Industrial policy and the policy toolkit

From grants to stakes. CHIPS (2022) gave about $39B of manufacturing incentives and $11B of R&D money [src-20220511-6efadf]. Now the government also takes equity: Commerce turned Intel's grants into a 9.9% stake ($8.9B), and Fortune counts about $26.7B of US government equity across roughly 30 deals, including $400M in MP Materials [src-20260726-39275b]. A "national champion" can get the state as shareholder and anchor buyer; a competitor it picks can crowd you out [inference].

CHIPS is ending. FY2026 was the last year of appropriated CHIPS manufacturing money, and the 35% 48D credit covers only projects started by December 31, 2026 [src-20260701-78183b]. Chip-adjacent materials startups (Periodic, CuspAI, Orbital) sell into an industry whose subsidy flow peaks now [inference].

IRA to OBBBA. The July 2025 law cut back much of the IRA but kept the 45X production credit; most phase-downs now begin in 2030, and products with "material assistance" from a prohibited foreign entity get nothing [src-20250608-08074e]. **Bonus depreciation is permanent:** a $2M machine can be deducted in year one [src-20260114-2075f7][src-20260101-08d847]. This helps profitable incumbents most and makes their robots cheaper, narrowing the startups' edge [inference].

Tool	What it is	What it does for producers	Source
NDAA	Yearly defense policy law	FY26: multiyear munitions (Sec. 804(b)); Industrial Base Fund purchase commitments up to 10 years with advance payments (Sec. 867); spare-parts review (Sec. 1842)	[src-20251218-f0d6ef]
DPA Title III	Lets the government pay to expand domestic capacity	23 awards, $820.5M since Jan 2025, mostly rare earths and rocket motors; none to tracked startups	[src-20251215-98f650][src-20261019-68fd3d]
OSC	Pentagon lender (2022) for 34 critical technology areas	$620M to Vulcan; $97.8M FY26 funding can back up to $4.4B of loans	[src-20251103-ee9b27][src-20260908-f6558e]
CHIPS Act	$52.7B for chip fabs, research and workforce	Co-funded Vulcan with a $50M incentive	[src-20230928-c47558][src-20251103-ee9b27]
IRA 45X	Per-unit credit for US-made energy components; minerals earn 10% of production cost	Minerals credit phases out 2031-2034; bars Chinese-linked supply chains	[src-20251001-45a0c1]
Section 232	National-security tariffs	50/25/10% metals tiers	[src-20260402-1b5d6c]
Buy American / Specialty Metals	DoD must buy US-made; specialty metals melted in the US	Protects domestic demand	[src-20251215-98f650]

What this means [inference]. Policy is strongest for metals and minerals; for machined parts it works mainly through prime contracts and the new multiyear authorities.

2.4 The government as lender and investor: OSC, EDF and DPA Title III

- **OSC (Defense).** June 2026 conditional loans of $725M to Energy Fuels and $500M to Phoenix Tailings [src-20260618-184e0c]; earlier $150M to MP Materials [src-20260807-110c13]. A new program lends $500M-$1B per private fund; applications due November 1, 2026; the FY27 request includes about $20B for OSC [src-20260831-4dd3b7]. An OSC official: without germanium, gallium and rare earths the buildout "is a pipe dream" [src-20260618-184e0c]. OSC debt can replace much of a heavy producer's equity, and the fund-finance window is worth a look for SPC itself [inference].
- **EDF (Energy; formerly the Loan Programs Office).** More than $289B of loan authority [src-20260116-27c00a][src-20251217-51eb2f], going to nuclear, grid, fossil and minerals, e.g. $17.5B conditional for Westinghouse AP1000 components; it wants offtakes, proven technology and US content [src-20260804-05a6a5]. Fits late-stage plants, but a reactor buildout means years of demand for forgings, castings and precision parts [inference].
- **DPA Title III and the Industrial Base Fund.** Grants and purchase commitments, plus DLA stockpile orders for magnets; USA Rare Earth got a $277M grant plus a $1.3B loan [src-20260807-110c13].
- **Mineral executive orders.** March 2025 speeds permits and directs DPA and DFC financing to minerals [src-20250320-91988a]; April 2025 opens seabed minerals [src-20250424-a310b3]. A 3-year permit instead of 10 changes value more than any grant [inference].

2.5 Capex, construction, rates and how plants get financed

- **Construction is cooling.** Manufacturing construction spending was $202.4B annualized in December 2025, almost 16% below its August 2024 peak, and down 21.9% year on year by May 2026; data centers are about 35% of contractor backlogs and compete for electricians and switchgear [src-20260302-477c5f][src-20260708-da5010].
- **Activity is strong.** ISM PMI 54.5 in September 2026, the ninth month above 50; new orders 55.3, prices paid 77.9 [src-20261008-89df84].
- **Capex before contracts. Claim (Power):** "no one's going to give you a production contract unless you've already started executing"; capex runs about 2 years ahead of revenue, and only customer prepayment or long-term infrastructure credit has worked [src-20260921-876080].
- **Debt for machines, equity for software. Claim (Mehr, Machina):** "don't fund physical hardware with equity"; machine loans cost about 2-3% over prime, 8-9% good to 12-14% risky [src-20260220-0d5345][src-20260320-1361e2][src-20260430-30ea09]. **Claim (Belosic):** lenders fund machines, not the software team [src-20260519-2d6ceb], and custom lines (anodizing, casting) are "unfinanceable" [src-20260624-3dc6f7]. Who funded what: section 10.
- **What this means [inference].** Parts demand is good; new plants are costlier. This favors capital-light models (brownfield robotics, roll-ups like Amca) over greenfield megafactories.

2.6 Defense and munitions budgets

- **The FY27 request.** CNAS counts $49.5B for 10 key munitions, about 8.5x the FY16-26 average of $5.8B. But about 82% depends on reconciliation, lead times have not improved for 6 of 10 munitions, the Iran war used more than half of US high-end interceptors, and framework deals with primes (Lockheed up to $9B, Northrop over $2B) are mostly unconverted; CNAS calls it "necessary but insufficient" [src-20261007-c6f64e]. CRS/CBO estimate a $21.7B replacement bill [src-20261005-b7e08b].
- **Primes sign capacity deals with new producers.** Lockheed's 7-year deal puts Hadrian cells inside its missile plant from early 2027 [src-20261014-015327]; the Navy put $900M beside

$1.5B+ of Hadrian money for submarine-part plants [src-20260320-67ba4c]; Divergent builds Tomahawk structures for RTX [src-20260601-976843]; the Army chose Vulcan magnets for a 1M-drones-a-year program [src-20260923-c53c99]. **Claim (Power):** labor hours per submarine rose from about 13M to over 70M [src-20260610-779f8b].

- **Navy on record.** “I do not believe we have a technology problem, we have an adoption problem”; the nuclear supplier base shrank from 17,000+ to about 5,000 [src-20250319-f21bec].
- **Demand is lumpy and orders lag budgets. Claim (Power):** the government “may not buy missiles next year” [src-20260908-65fab6]. Carpenter’s defense bookings fell because customers “are not receiving funds” [src-20261022-de5e6c].
- **What this means [inference].** The biggest demand driver, but back-loaded and political. Ask candidates how much backlog is funded versus framework-only.

2.7 Labor: the binding constraint, and why cheap labor is not the point

- **The gap.** 3.8M manufacturing jobs to fill from 2024 to 2033, 1.9M possibly unfilled [src-20240416-b69154]; 522,000 open jobs in August 2026 versus 432,000 pre-pandemic [src-20261013-a7c995]; DoD needs at least 122,000 more critical workers by 2028 [src-20250101-af28c1]. 2.3M technician openings by 2030 are counted separately and exclude machinists, so do not add them [src-20260910-a2cd42]. Welders needed by 2029: 320,500 [src-20261023-57ad95].
- **Owner-operator claims.** Hadrian trains operators in 30-40 days [src-20260926-70087f]; SendCutSend hired 200 welders in a year, mostly with no factory experience [src-20260525-4a12ab].
- **The correction. Claim (Drew Baglino):** in a new automated plant “the labor differential is less than 10% of COGS”; China wins because its supply chain sits within a 3-hour drive [src-20260513-bdc044].
- **What this means [inference].** The pitch is “we deliver when your job shop can’t hire”, not cheaper labor.

2.8 Power, permits and build time

- **Claim (Belosic):** “We need 100 times more power”; transformers and permits are delayed [src-20260519-5cfe14]. Permits for identical equipment take “3 weeks, 3 months, or 3 years” by town [src-20260813-1fe8a2]; Vulcan’s CEO: a multi-year permit means “running out of cash” [src-20250429-6f9155].
- **Claim (Mariana CEO):** a mine or refinery can take about 5 years to build and 3-5 to ramp [src-20260513-bdc044]; Chinese refineries commission in about 6 months versus 2-4 years in the West [src-20250723-3f3ef8].

2.9 Venture climate and how investors see it

- US VC hit $515.8B through Q3 2026, 82.7% to AI; exits are stalled [src-20261008-8f7d66]. Defense-specific startups raised a record $2.5B in Q3; seed deals fell to 41, the fewest since 2016 [src-20261006-0f220a]; defense-tech money is concentrating in fewer winners [src-20260908-abd0c7]. **Claim (Construct Capital):** later-stage industrial funding grew about 10x in 4 years [src-20260702-d79e75].
- **Margins are lower than software:** Proto Labs 44.5% [src-20260206-8bb619]; service centers 18-32% [src-20260729-407fd7][src-20261021-16bb64]; roll-ups “effectively … 30%” [src-20241001-28c2aa]. Bull: **Claim (Power):** 65-80% utilization versus about 10% at primes [src-20260610-779f8b]. Bear: “better suited to debt than VC” [src-20250709-43486e]; failures in section 8. No exit of an AI-native producer above $100M (Appendix B).
- **What this means [inference].** Two openings: better seed/A prices for producers outside the hype labels, and a risk that the best producers are priced as AI companies. The test for SPC: can

a software-run plant beat job-shop margins and turn prototype wins into multi-year orders?

2.10 Science policy: the Genesis Mission, DOE, NSF and the national labs

- Genesis (EO 14363, November 2025) directs DOE to build an AI platform linking lab supercomputers, datasets and robotic labs; priorities include advanced manufacturing and critical materials [src-20251124-b97a20]. October 8, 2026: 12 Phase II awards worth $159M; 297 first-year projects [src-20261008-452806]. NSF spends more than $700M a year on AI research [src-20261019-5257f3]. Labs already work with startups (Sandia with Radical AI) [src-20260929-dfb65a].
- **What it means [inference].** Genesis is both funder and competitor: free compute and test beds, but also free public models that weaken a “proprietary model” moat. Startups win on what they make and qualify.

2.11 More macro facts a newcomer should know

- 54.9% of manufacturers cite workforce as a top challenge [src-20260914-b4d55c]. Fully loaded US factory labor is about $46.30 an hour, while a robot fell from $46,000 (2010) to $23,000 (2022) (newsletter) [src-20260826-14e4bb].
- Precedent: in the 1950s the Pentagon paid $239M (about $2.8B today) for 10 giant forging presses and leased them to industry [src-20240821-44ed1d].

2.12 Market size: overview

No layer has a clean, free TAM. Each chapter gives its own numbers.

Layer	Best free number	Gap
All US manufacturing	$2.90T value added [src-20261013-a7c995]	n/a
Metal distribution	About $200B across about 9,400 sites; top 50 hold about half [src-20251201-4b732d] (med)	Excludes refining
Metal refining	Public money committed to minerals: $65B [src-20260716-6a06e4]	No TAM
Machine shops	17,156 sites, $14.23B payroll [src-20250626-23f6d9]; precision parts about $71B by 2026 [src-20251024-2e5383] (med)	Counts, one memo
Fabrication	14,533 structural/architectural metal sites, $28.26B payroll [src-20250626-23f6d9]	No revenue
Factory software	Robots installed worldwide $18.6B (2025) [src-20260924-28b59b]	Robots, not software

2.13 Data loops: what the sources say for and against

- **For: the data does not exist unless you make it.** Periodic: published values for one property “spanned many orders of magnitude”, and negative results go unpublished [src-20260403-ba156a][src-20260106-62dba1]. **Claim (Power):** manufacturing data is “never on the internet” [src-20260908-65fab6]. Mechanisms: **Claim (SendCutSend):** “the more jobs that we take in, the smarter we get” [src-20260626-541ac8]; **Claim (Machina):** a SpaceX part went from 25 trials to 4-5 [src-20260926-70087f].
- **Against: siloed, slow and fragile.** Plant data sits in separate layers [src-20250409-8f07e3]; “models are largely good enough”, the bottleneck is data and adoption [src-20260922-94fcb0];

some Periodic workflows are “still like 90% human” [src-20261008-83dc1b]; a pilot fails if “one false detection” makes operators switch it off [src-20261011-5db652]; Argonne publishes its robot-lab data openly [src-20250217-059784].

- **Incumbents use AI too. Claim (Foxconn):** +15% labor productivity, +3% first-pass yield [src-20261022-89d710]. a16z: US firms are “vertically integrating by necessity, not strategy”, while incumbents build “robotic arm farms” for their own data [src-20251225-725014, 6:38, 19:17].
- **What this means [inference].** Running the plant creates data others lack, but nobody yet shows cost falling with volume.

2.14 Macro scorecard by industry [inference]

Force	Raw Materials (A / B)	Manufacturing	Specialty Chemicals	Advanced Materials	Software & Robotics
OSC/EDF/DPA money	Strong + (most to B)	+ (via defense)	Mild +	+ (DOE battery awards)	0
China export controls	Strong + (B most)	Input-cost −	+ (pharma reshoring)	+	0
Section 232 tariffs	A mild +; B strong + (makers), − (buyers)	Mixed	+ (pharma; generics excluded)	0	0
Rates 3.75–4.00%	−	Mild −	−	−	Mild −
Munitions budget	+ (B: magnets, alloys)	Strong +	Mild +	0	Mild +
Labor shortage	Mild −	Strong +	Mild +	Mild +	Strong +
VC concentration	−	Hype pricing	Mild −	Hype pricing (labs)	−

2.15 What would break “why now”

- Tariffs reversed or limited by courts would hit metals and fabrication first [inference].
- A flat defense budget or delayed munitions buys would strand plants built ahead of contracts (2.5, 2.6).
- Higher rates; a lasting US–China minerals deal that erases domestic premiums [inference].
- Spare capacity: the average CNC spindle cuts metal 34% of the time versus a 70% target [src-20250120-419768] (med).

2.16 Listen & watch

Free episodes and videos for this chapter (from listening list v2; links checked Oct 27, 2026). ★ = top pick. Full list: Appendix D.

1. ★ **“China Has Mass. Can America Catch Up?”**, a16z Podcast, Sep 29, 2025, length not listed. ▶ Open link — *Why:* Anduril’s and Hadrian’s CEOs explain the munitions gap: war games run out in about a week, restocking takes 2–3 years. Repo: src-20250929-452dcb.
2. **“How LA Is Becoming America’s New Industrial Tech Hub”**, Bloomberg *Odd Lots*, Sep 21, 2026, 34 min. ▶ Open link — *Why:* the best mainstream framing; the bottleneck is factory software and feedback, not robots. Repo: src-20260921-a51ab2.
3. **“Can the U.S. Still Build Things?” (#34)**, *The Drone Ultimatum*, with Miles Arnone (Re:Build), about 95 min. ▶ Open link — *Why:* the case for buying and upgrading old shops instead of building megafactories. Repo: src-20261022-f423ef.

3. Raw Materials (Minerals & Metals)

> **Our view: Raw Materials** *(analyst judgment, not fact [inference])* - **Stance:** Selective. Back refining and magnets (Stage B) where a government or defense buyer is already paying, and watch exploration (Stage A). - **Conviction:** medium. Policy pull is the strongest of any industry (2.14), but profits depend on prices, which a startup cannot control. - **Why:** (1) only the owner of a non-Chinese plant can sign the offtake that OSC and EDF lenders want [src-20260804-05a6a5]; (2) the data loop scores 2 where the process runs continuously (Mariana); (3) startups win on small deposits, tailings and magnet steps that majors ignore (3.14). - **Strongest counter:** MP, the largest US name, still loses money [src-20260201-0b4d7a]. A Chinese price cut or a US–China deal could strand subsidized plants. - **What would change our mind:** price floors renewed without subsidy; Mariana hits nameplate on schedule; OSC fund-finance commitments close (expected by March 2027). - **Best placed / watch:** Vulcan (fully integrated), Mariana (fully integrated); watch Earth AI (hybrid) and Phoenix Tailings (context). - **Good SPC fit:** Series A–B; a plant financed mostly by government debt, with venture money paying for software and ramp; buyer is DoD, a prime or an automaker with an offtake.

What this industry is. Raw Materials covers everything from finding ore to delivering a finished metal product: **Stage A, finding and mining** (exploration, mining and concentrating ore) and **Stage B, refining and making metal** (refining to pure oxides, carbonates or cathode, then smelting, alloys, magnets and powders). It sits at the bottom of every supply chain in this primer. Metal distribution (service centers that buy mill metal and cut it to size) now sits in Manufacturing (chapter 4) as a supplier to parts makers.

3.1 Shared macro: China’s refining grip, the critical-minerals orders and government lending

- **China owns the middle of the chain.** It leads refining for 19 of 20 strategic energy minerals (about 70% share) [src-20250521-c62d29], did about 99% of heavy rare-earth processing until 2023 [src-20250414-50f023], makes about 60% of refined germanium [src-20261027-f82d55] and over 90% of rare-earth magnets (US under 1%) [src-20250429-6f9155]. Rare earths go into the F-35, Virginia- and Columbia-class submarines, Tomahawk and JDAM [src-20250414-50f023]. **Claim (Power):** about 150 of 1,000 components in a Navy missile come from China [src-20260610-779f8b].
- **China uses the leverage.** Licensing on 7 heavy rare earths and magnets since April 2025 (still in force); broader October 2025 controls suspended to **November 10, 2026** (the US says the truce runs to January 10, 2027); the gallium, germanium and antimony ban on the US suspended only to **November 27, 2026** [src-20261019-854e22][src-20261027-f82d55]. Rare-earth exports fell 9.6% year on year in January–September 2026 [src-20261019-854e22]. Strature puts a 63% probability on a short roll-over [src-20261016-83adeb] (med).
- **Prices show stress.** Germanium hit a record ¥28,500/kg on October 26, 2026, up about 110% year to date, with a 78.6% US-warehouse premium [src-20261027-f82d55]. In 2025 antimony rose 144%, while lithium fell 24% and nickel 11% [src-20260130-ebdd5a].
- **Not enough capacity is coming.** About $6.5 trillion a year of output depends on concentrated minerals; planned non-Chinese magnet capacity covers about a third of 2035 demand [src-20261014-b9d647].
- **Government lending and orders.** OSC, EDF, DPA and equity stakes (2.3-2.4) are mostly minerals deals; $65B of public money was committed to minerals in 2025 [src-20260716-6a06e4]. The 2025 executive orders speed permits [src-20250320-91988a][src-20250424-a310b3]; a new president can reverse them, but granted permits stay granted [inference].

- **What it means [inference].** A US producer of rare-earth metals, magnets, gallium or germanium has the government as buyer of last resort. But if China floods the market after a deal, prices collapse, so producers need floor-price offtakes (the MP Materials–DoD template) [src-20260726-39275b]. “Designing out” scarce elements is the strongest commercial case for materials AI (chapter 6).

3.2 Stage A — Finding and mining: problem, tailwinds and headwinds

The problem. US nonfuel mines produced $112B in 2025, but metals were only $38.1B [src-20260130-ebdd5a]. The US is 100% import-reliant for 16 mineral commodities and over 50% for 54, including 67% of rare earths and 57% of refined copper [src-20260206-04fe3e]. **Claim (Earth AI):** industry discovery success is about 0.5% per project [src-20250127-c28430].

Tailwinds, ranked [ranking is inference].

Rank	Tailwind	Date / size	Structural or reversible?
1	**Permitting orders**	March and April 2025 [src-20250320-91988a][src-20250424-a310b3]	Reversible; granted permits stay
2	**Defense, grid and data-center demand**	Rare earths in weapons [src-20250414-50f023]; an AI campus can need up to 50,000 t of copper [src-20261007-a642ea]	Defense structural; AI capex could slow [src-20261015-c425a9]
3	**Federal lending and equity**	See 3.1 and 2.4	Durable this administration
4	**45X tax credit**	10% of production cost, phased out 2031–34 [src-20251001-45a0c1][src-20250608-08074e]	Legislated, with a sunset
5	**Stockpiling**	Processed-mineral imports rose to $185B from $77B in 2025 [src-20260130-ebdd5a]	Temporary

Headwinds, at equal weight. 1. **Commodity prices swing** (lithium −24%, nickel −11% in 2025) [src-20260130-ebdd5a]. 2. **Permits and time.** About 5 years to build and 3–5 to ramp after permits (**Claim, Mariana CEO**) [src-20260513-bdc044]; detail in 3.14. 3. **Geology decides.** Ore grade, not software, sets most of the cost [inference]. **Claim (Earth AI’s CEO):** about 40 deposits are found a year and about a fifth become mines; about 650 more mines are needed by 2040 for five battery/magnet minerals, yet only a handful opened in 2023-25; about $300M of industry spend per discovery [src-20260428-aa5557, 45:54–47:04][src-20251001-cf441c, 2:10–4:46]. 4. **Rates.** Mines are the most rate-sensitive assets here [src-20260916-f0c016][inference]. 5. **A US–China deal** takes the urgency out (3.1).

3.3 Stage B — Refining and making metal: problem, tailwinds and headwinds

The problem. Chinese refineries commission in about 6 months versus 2–4 years in the West [src-20250723-3f3ef8]. China produces 56% of world castings, and the Pentagon relies on one US foundry for large titanium parts (General Catalyst thesis) [src-20240202-8c6d3b]. Steel mill products are up about 86% and copper wire about 91.4% since February 2020 [src-20260302-477c5f]; Reliance calls aluminum “historically elevated largely due to” tariffs [src-20260728-ae7dea].

Tailwinds, ranked [ranking is inference].

Rank	Tailwind	Date / size	Structural or reversible?
1	**China's export controls** make non-Chinese supply a security good	See 3.1	Structural in direction; timing swings with any deal
2	**Defense and aerospace demand**	FY27 munitions request (2.6); Army drone magnets from Vulcan [src-20260923-c53c99]; Carpenter aero bookings +49% y/y [src-20261022-de5e6c]	Structural 5–10 years; budgets swing
3	**Section 232 tariffs**	50/25/10% tiers since April 2026 (2.2) [src-20260402-1b5d6c][src-20260602-b995e2]	Proclamation; can reverse
4	**Money for magnets and specialty metals**	$1.4B Vulcan/ReElement package [src-20251103-ee9b27]; NDAA Sec. 867 [src-20251218-f0d6ef]	Durable in law; funding annual
5	**Buy American / specialty metals**	DoD titanium, nickel, some steels melted in the US [src-20251215-98f650]	Statute
6	**Data-center and grid build-out**	Up to 200,000 t of steel per AI campus [src-20261007-a642ea]; AI capex 3.63% of GDP, above the railroad boom [src-20261015-c425a9]	Reversible if AI capex slows

Headwinds, at equal weight. 1. **China can crash prices on purpose** (3.1). 2. **Tariffs cut both ways:** Nucor's margin rose to 20% from 14%, but its own fabrication unit reported "margin compression" [src-20260812-77d6f5]. Incumbents are winning this cycle [inference]. 3. **The equipment is Chinese too** (**Claim, Vulcan CEO**) [src-20250429-6f9155]. 4. **Orders lag the budget** (Carpenter, 2.6) [src-20261022-de5e6c]. 5. **Plants are capital-heavy:** Vulcan's plant rests on a $1.4B package [src-20251103-ee9b27]; custom lines are "unfinanceable" (**Claim, Belosic**) [src-20260624-3dc6f7]. 6. **The state picks winners** [inference].

3.4 Why an AI-enabled, vertically integrated company is the answer here

- **Stage A:** AI narrows where to drill; owning the claims and rigs keeps a discovery's upside. **Where it breaks [inference]:** a find is worth little until drilled out, and most are sold to miners.
- **Stage B:** the scarce thing is a permitted, financed plant outside China. Only the owner of output can sign the offtake lenders want [src-20260804-05a6a5]. AI attacks build and ramp time: **Claim (Mariana):** software halves 5–10-year timelines, and its first plant runs about 50% autonomously with agents retuning every 15 minutes [src-20260803-30e6d5][src-20260623-d1790b]. **Claim (Vulcan):** its pilot opened March 31, 2025 and was qualifying magnets within months [src-20250811-e7cb9a].
- **Where it breaks [inference].** Ore grade and price cycles decide most of the economics; a refinery is one giant bet, so venture equity is the wrong money for most of it.

3.5 What would have to be true, and signals to watch

- A non-Chinese price floor holds after a US–China deal. **Watch:** November 10 and 27, 2026; DoD price-floor deals.
- AI shortens build and ramp. **Watch:** Mariana at nameplate; recovery and cost per ton.
- Government debt replaces most equity. **Watch:** OSC fund-finance commitments by March 10, 2027 [src-20260831-4dd3b7].

- An AI explorer defines a resource. **Watch:** Earth AI at Fontenoy or Willow Glen [src-20241121-ac9f37].
- US magnet cost gets close enough to China that offtakes renew without subsidy. **Watch:** Vulcan's first production.
- Section 232 survives and FY27 defense money flows.

3.6 Macro in more depth

- **Protected versus open markets.** DFARS bars defense buyers from a "foreign entity of concern", so price against China does not decide the sale [src-20250429-6f9155]; ITAR keeps defense work onshore. **Claim (Mehr):** "if you go head-to-head with China … you're losing", so Machina starts in defense [src-20260622-7846f9]. In open markets SendCutSend can "meet or beat China" on only about 20-25% of its work [src-20260519-2d6ceb], and Taiwan is 20-30% cheaper even with tariffs [src-20250709-43486e]. Commercial buyers do pay some insurance: a few hundred dollars of parts can stop an $80,000 car (**Claim, Vulcan CEO**) [src-20250429-6f9155].
- **Tariffs, net.** Section 232 helps metal makers and hurts metal buyers, including fabrication startups; the 10% vs 25% gap rewards suppliers who can prove where metal was melted [src-20260402-1b5d6c][inference]. ENR notes an energy-price spike tied to the Iran conflict [src-20260708-da5010].
- **Data-center demand.** A legacy fabricator building generator enclosures for hyperscalers has "backlogs working on 2027" (**Claim**) [src-20261022-4234b3]; most hyperscalers have near-zero free cash flow [src-20261015-c425a9].

3.7 How the industry and its supply chain work

- **Stage A:** explorers find and drill out a deposit, usually selling it to a miner, which mines and concentrates the ore.
- **Stage B:** concentrate is refined to an oxide, carbonate or cathode, then smelted into alloys, magnets, powders and mill products for service centers and parts makers (chapter 4) or chemical and materials plants (chapters 5-6).
- **Buyers and prices.** Miners, metal and magnet makers, battery makers, primes, the DoD stockpile and traders. Prices are set globally or in China, so a US producer is a price-taker without a floor-price offtake [inference]. Incumbents: Nucor, Carpenter, MP Materials. Candidates: Hertha, Cyclic, Mass Magnetics, Duranium.
- **Market size.** No free source sizes US refining or exploration; public money committed to minerals: $65B in 2025 [src-20260716-6a06e4].

3.8 Where AI and automation help

Stage A: AI targeting and ore-body modeling. Stage B: AI agents tuning a refinery, recipe tuning for magnets and alloys, faster engineering and permitting documents (MarianaOS) [src-20260803-30e6d5].

3.9 Right to win, unit economics and data loop

Right to win: Stage A, owned claims and drills; Stage B, a permitted plant outside China plus software that shortens build and ramp. **Unit economics:** cost per ton, recovery and cost per discovery are undisclosed for every startup [gap].

Company	Loop score (0–3; 2.13)	How it works	Missing
Mariana	2	Agents retune the refinery every 15 minutes [src-20260623-d1790b]	Recovery or cost per ton
Vulcan	1	Test data feeds recipe models [src-20250811-e7cb9a]	Any number

3.10 Companies, grouped by label

Stage A — Finding and mining

> **Our view: Stage A, exploration** *(analyst judgment)* — **Watch, not invest yet.** **Conviction: medium. Why:** AI lowers the cost of finding a deposit, but a find earns nothing until a mine is built, and mines take 15.7–19.1 years from discovery (3.14). **Counter:** cheap drilling plus owned claims could let an explorer sell deposits early to miners, a real business. **Change our mind:** a formal resource estimate at Fontenoy or Willow Glen, or a deposit sale to a major. **Watch:** Earth AI (hybrid). **Good fit:** owns claims and rigs, sells deposits to majors, buyer lined up before drilling out.

Earth AI (mineral exploration; hybrid; owns claims, drills, sells deposits)

- **What it does.** Its own AI targeting model (a deep neural net trained on ~400M sites, “not generative AI”) picks targets in poorly explored Australian ground; it drills them with its own rigs and is building its own assay labs to cut results from five months to days [src-20250127-c28430] [src-20260429-1bdb70][src-20260428-aa5557, 62:00]. **Claim:** about $100 a meter drilling and up to 80% lower exploration cost [src-20250127-c28430]; cost per discovery “under 1 million” vs ~$300M industry [src-20251001-cf441c, 27:22].
- **Track record (claim, CEO Roman Teslyuk).** Of 5 drilled prospects, 4 became discoveries (“grade in core”, not tonnage), the update of the earlier 75% figure; sizing results were due “in the next six to nine months” (said April 2026) [src-20260428-aa5557, 52:43–53:32]. Reported finds: Fontenoy palladium-platinum-nickel across about 2 km² [src-20241121-ac9f37]; Willow Glen gold, seven intercepts at about 1.14 g/t [src-20241220-f860e9]; Kooranjie and Elkedra rock chips (tungsten, cobalt, copper, indium) [src-20250326-5929b8][src-20250601-b499c3].
- **Model.** Owns most of its ground (3.1M acres, NSW’s biggest landholder, under 40 staff) [src-20250326-5929b8][src-20251001-cf441c, 13:51]; took 80% of Fontenoy for A$4.5M of work [src-20241121-ac9f37]. Plans to prove deposits and sell them at about 5% of the metal in the ground, keeping a stake or royalty, and runs ~30 projects like a venture portfolio [src-20251001-cf441c]. It integrated because 80 software pilots with miners did not recur [src-20251001-cf441c, 24:26] [src-20260428-aa5557, 41:21].
- **Funding.** $20M Series B, January 2025 (Tamarack Global, Cantos) [src-20250127-c28430].
- **Bear case.** No formal resource estimate (e.g. JORC) published [gap]; the success rate is self-defined on a small sample [src-20260428-aa5557, 52:50]; assets are in Australia; the exit is a project sale, so most mine value goes to the builder [src-20251001-cf441c]; competition from better-funded KoBold, plus “maybe 50 new startups” [src-20260428-aa5557, 56:14].
- **Next step [inference].** Drill results at Fontenoy, a resource-estimate timeline, and buyers spoken to.

Mariana Minerals (mining and refining; fully integrated; fits)

- **What it does.** Builds and runs mines and refineries (Copper One, Lithium One) on its own software; “we do not sell software” [src-20260803-30e6d5][src-20260623-d1790b]. Selling

software to miners is a “death spiral” of pilots, says CEO Turner Caldwell [src-20250723-3f3ef8, 21:52].

- **Operations (claims).** Restarted a Utah copper mine in May 2026, about 50% autonomous (autonomy bought in from Sandvik and Pronto), making 99.99% copper; the prior operator shut it partly for lack of haul-truck drivers [src-20260623-d1790b][src-20260502-21f16a]. Targets 50,000 t a year by 2030 versus about 2,500 t before [src-20261011-e2426c]; 30% lower refining and 40-50% lower mining cost [src-20260502-21f16a]; 10 projects in 10 years, two today; East Texas lithium to an unnamed battery maker [src-20260513-dd56eb][src-20261024-c53370].
- **Money.** $310M Series B (Khosla), about $400M including project money; most mine capex comes from project investors [src-20260803-30e6d5][src-20261024-c53370]. A $1.5B valuation appears only in Harmonic (low). About 250 staff, 20% in software [src-20260623-d1790b].
- **What this means [inference].** One of two names with a credible data loop; the check size is above SPC’s. Mariana’s refinery is also its Stage B entry.

Stage B — Refining and making metal

> **Our view: Stage B, refining and metal** *(analyst judgment)* — **Attractive now, selectively. Conviction: medium. Why:** the scarce thing is a permitted, working plant outside China; DoD and OSC money flows here first (13, conclusion 5); Vulcan’s pilot output is sold out 12 months ahead (**claim**) [src-20250811-e7cb9a]. **Counter:** a refinery is one giant bet better suited to debt, and Ganfeng lost over 75% of its value with lithium prices [src-20250529-3a588b]. **Change our mind:** offtakes that renew without subsidy (bull); a court overturning Section 232 (bear). **Best placed:** Vulcan, Mariana (fully integrated). **Good fit:** pilot already shipping to a qualified defense buyer; government debt for plant two.

Vulcan Elements (magnets; fully integrated; fits)

- **What it makes:** rare-earth magnets; a 10-ton pilot in Durham and a 10,000-ton plant planned [src-20251103-ee9b27].
- **Money:** $65M Series A (Altimeter) [src-20261010-8e6ec0]. A $1.4B package: $620M loan from the Pentagon’s OSC, $50M Commerce Department equity, $550M private [src-20251103-ee9b27]. Form D filing shows $430.7M sold [src-20260402-c8e4dc].

Context names (not profiled): Phoenix Tailings (rare-earth metal and magnet feedstock; $500M conditional OSC loan, facility due 2028) [src-20260618-184e0c]; MP Materials (incumbent; $400M DoD equity) [src-20260726-39275b]; Energy Fuels ($725M conditional OSC loan) [src-20260618-184e0c]; ReElement (Vulcan’s refining partner; $80M OSC loan) [src-20251103-ee9b27]; Hertha, Mass Magnetics, Duranium (candidates). Cyclic (candidate, not sourced this run).

3.11 Bear case, at equal weight

- **Both stages:** a US–China deal and Chinese price-dumping can make every Western project uneconomic overnight (3.1); the state picks champions with equity (MP Materials) [src-20260726-39275b].
- **Stage A:** commodity cycles (lithium −24%, nickel −11% in 2025) [src-20260130-ebdd5a]; ore grade, not software, sets most of the cost [inference]; Earth AI has no published resource estimate (3.10).
- **Stage B:** incumbents are winning this cycle (Nucor margins) [src-20260812-77d6f5]; tariffs can be reversed and squeeze metal buyers; magnet economics depend on subsidy and offtake, and equipment comes from China (**Claim, Vulcan CEO**) [src-20250429-6f9155]; defense orders lag

budgets (Carpenter) [src-20261022-de5e6c]; Mariana has disclosed no recovery or cost-per-ton number [src-20260623-d1790b].

3.12 Key sources

USGS [src-20260206-04fe3e][src-20260130-ebdd5a]; IEA [src-20250521-c62d29][src-20261014-b9d647]; MOFCOM and truce dates [src-20261019-854e22]; Ga/Ge/Sb [src-20261027-f82d55]; CSIS [src-20250414-50f023]; OSC [src-20260618-184e0c][src-20260831-4dd3b7]; EDF [src-20260804-05a6a5]; Section 232 [src-20260402-1b5d6c]; Vulcan/OSC [src-20251103-ee9b27]; Earth AI [src-20250127-c28430][src-20251001-cf441c].

3.13 Listen & watch

Full list: Appendix D.

Stage A — Finding and mining

1. ★ **“Critical Minerals: Mining for the Industrial Future”**, a16z Podcast, Jul 23, 2025, length not listed. ▶ Open link — *Why:* Mariana’s Turner Caldwell on owning mines and refineries and using software to cut build time and cost. Repo: src-20250723-3f3ef8.
2. **“Inside Earth AI: Cutting the Cost of Critical Mineral Discovery” (S2 E57)**, Rare Earth Exchanges, Feb 26, 2026, 33 min. ▶ Open link — *Why:* founder Roman Teslyuk on training on 50 years of Australian drilling archives, and why few new deposits are found each year (his figures). Repo: src-20251001-cf441c.
3. **“From Y Combinator to mining, ft Earth AI’s Roman Teslyuk”**, YouTube, Apr 28, 2026, 66 min. ▶ Open link — *Why:* the longer Earth AI founder story; pair with the primer’s bear case (no resource estimate yet). **Not in the repo (new).**
4. **“Critical Minerals: What Makes Mining Tech VC-Backable”**, *Deep Tech Catalyst*, with Jun Ku (Main Sequence), length not listed. ▶ Open link — *Why:* an investor on how non-dilutive government money makes heavy-capex mining fundable. Repo: src-20261022-cd6c5c.

Stage B — Refining and making metal

1. ★ **“How The Rarest Company On Earth Raised $75M”**, *Sourcery*, Aug 11, 2025, length not listed. ▶ Open link — *Why:* Vulcan’s early demand: pilot plant opened Mar 2025 and was qualifying magnets with customers within four months (company claim). Repo: src-20250811-e7cb9a.
2. **“‘Supply Chains Win Wars’: Rare Earth Magnets with John Maslin”**, *Building the Base*, Apr 29, 2025, length not listed. ▶ Open link — *Why:* the magnet macro in one sitting; the gap is processing and metal-making, not ore. Repo: src-20250429-6f9155.
3. **“Energy, Minerals, and the Physical Stack Behind AI”**, a16z Podcast, May 13, 2026, length not listed. ▶ Open link — *Why:* Mariana’s refinery and plant software (the data loop inside owned assets). Repo: src-20260513-bdc044.
4. **“Advancing US critical minerals with software and autonomy”**, Benchmark Mineral Intelligence, Jun 23, 2026, 23 min. ▶ Open link — *Why:* how industry analysts see AI-run refineries. Repo: src-20260623-6dc76b.
5. *(Quick watch)* **“The Future is Metal – Mariana Minerals”**, a16z, Aug 5, 2026, 9 min. ▶ Open link — *Why:* nine minutes on China’s 90%+ share of refining. Repo: src-20260805-b7e3c7.

3.14 Commercial traction and incumbents

*Status labels: **Revenue** = paying customers or product sales; **Pilot** = pilots, LOIs, government awards or samples; **Lab** = pre-revenue. Company statements are marked **Claim**.*

Who’s actually commercializing

Company	Sub-section	Status	Evidence
Mariana Minerals	Stage A/B (mines and refines)	Revenue-stage producer; buyers not disclosed	Restarted its Utah copper mine in May 2026, about 50% autonomous; **Claim:** 99.99% copper [src-20260623-d1790b][src-20260513-bdc044]. **Claim:** East Texas lithium goes to an unnamed battery maker, ramping next year [src-20261024-c53370]. No buyer contracts public.
Vulcan Elements	Stage B (magnets)	Pilot, with government buyers	Seven federal awards worth $5.7M, including an Army magnet IDIQ [src-20260710-6361ea][src-20250501-1885f8]. **Claim:** pilot output sold out 12 months ahead [src-20250811-e7cb9a].
Earth AI	Stage A (exploration)	Lab / pre-revenue	No published resource estimate [src-20250127-c28430][src-20241121-ac9f37].
Phoenix Tailings (context)	Stage B (rare-earth metal)	Pilot	$500M conditional loan; commercial facility due 2028 [src-20260618-184e0c].

Revenue vs lab. Real revenue today sits with incumbents in refining and magnets (MP Materials: $224.4M revenue in 2025, still a $85.9M net loss [src-20260201-0b4d7a]). Startup revenue is thin: Mariana is producing, Vulcan sells pilot volumes and federal work. AI exploration (Stage A) is pre-revenue by nature: a find earns nothing until a mine is built.

Incumbents. BHP, Rio Tinto and MP Materials. *What they do better:* balance sheets, permits in hand, offtake relationships and operating know-how; MP also has $400M of DoD equity [src-20260726-39275b]. They buy AI rather than build it: BHP Ventures backs AI exploration startups [src-20261022-04e57d]; Fleet Space lists Rio Tinto among the miners using its surveys [src-20261014-84a296]. *Where startups win [inference]:* small deposits and tailings the majors ignore, and refining or magnet steps where policy pays for a non-Chinese source.

Regulation and qualification. Permitting is the clock. A federal environmental impact statement (NEPA) took 4.5 years on average, 3.5 median, and a quarter took over 6 [src-20200612-38edca]. US mine permitting commonly takes 7–10 years versus about 2 in Australia and Canada; mines average 15.7 years from discovery to production [src-20240101-a81fa2], rising to 17.8 years for mines opened 2020–24 (US: 19.1) [src-20250101-3e6713]. Refinery build times: see 3.6 [src-20250723-3f3ef8]. Magnet buyers (defense, autos) then qualify the material, see 6.14.

5-year outlook (to 2031). *Stage B refining and magnets:* **possible**. Vulcan and Mariana have product and government pull. *Stage A AI exploration:* **unlikely**. Discovery-to-mine timelines run well past 2031. **Bear case, equal weight:** the subsidy-backed buyers (DoD, OSC loans) may not renew, loans are still conditional, and Chinese price cuts can strand US plants; MP is the largest US name and still loses money [src-20260201-0b4d7a].

4. Manufacturing

> **Our view: Manufacturing** *(analyst judgment, not fact [inference])* - **Stance:** Selective. Back capital-light owner-operators where speed and qualification are already paid for, and use the large names as benchmarks. - **Conviction:** medium. Demand is real and policy now buys orders, but price is not the wedge and no one shows unit cost falling with volume. - **Why:** (1) defense buyers pay for speed and capacity (section 0, finding 1); (2) SendCutSend has been profitable for 8 years [src-20260519-38c7ed], so the model can work; (3) the FY26 NDAA adds multiyear munitions and advance payments [src-20251218-f0d6ef]. - **Strongest counter:** at 1,000 parts an old shop is cheaper [src-20260908-ed94f0], and AI CAM tools are sold to every shop [src-20260908-f98d21]. - **What would change our mind:** a startup above 20% of a prime's rate-production spend at a lower price; published scrap or cycle-time curves improving with volume. - **Best placed / watch:** Senra and Freeform (fully integrated), Isembard (hybrid); benchmarks Hadrian, Divergent, SendCutSend. - **Good SPC fit:** Series B; owns the shop floor; already paid by primes or defense for speed; debt funds the machines.

What this industry is. Manufacturing turns metal (and some plastics and composites) into finished parts made to a customer's drawing. We use three sub-types, in chain order, and tag every company with one:

- **Metal Supply:** buying standard metal from mills and cutting it to size for parts makers (Nox Metals). These are suppliers to the parts makers below, not raw-materials producers.
- **Fabrication (sheet/structures):** cutting, bending and welding sheet metal and plate into brackets, enclosures, frames and structures. Lower tolerance, high volume of simple jobs, fast turnaround; margin depends on machine use and material cost.
- **Precision & Defense Hardware (machining, 3D printing, castings, composites):** tight-tolerance parts for aerospace, defense and medical, made by CNC machining, additive manufacturing (3D printing), casting or composites layup. Slower, needs certifications (AS9100, NADCAP) and a first-article inspection; higher margin.

Between fabrication and precision, the difference matters because fabrication competes on speed and price for many small orders, while precision competes on qualification and trust for long-lived programs. The macro below covers both; where it differs, we say which.

4.1 The problem today

- **The supplier base is small, old and shrinking** (2.1), and buyers have consolidated.
- **Parts are late.** An average aerospace part takes 8–12 weeks, and about 40% arrive late or with quality issues (med, single memo) [src-20251024-2e5383]. Castings take 9–10 months at a Raytheon supplier [src-20250708-2aa3ad].
- **Yet existing shops are underused:** machines cut metal 34% of the time against a 70% target [src-20250120-419768]. The shortage is people, programming and process, not machines [inference].
- **Labor.** Machinists: 344,900 jobs, $59,620 median pay, 30,400 openings a year from retirements [src-20260901-d504f9]. Welders: 437,700 jobs, 40,300 openings a year [src-20260901-3b0073].
- **Moving a part is slow by design:** first-article inspection repeats whenever design, material, process, tool or location changes [src-20190801-e81d7b].

4.2 Macro and tailwinds, ranked [ranking is inference]

Rank	Tailwind	Date / size	Structural or reversible?
1	**Munitions and shipbuilding demand**	FY27 request and prime deals (2.6)	Structural; budget timing reversible
2	**Multi-year buying in law**	FY26 NDAA Secs. 804(b), 867, 1842 [src-20251218-f0d6ef]	Statute, needs appropriations
3	**Tariffs on imported parts**	25% on goods containing steel, aluminum or copper [src-20260402-1b5d6c]	Reversible
4	**Reshoring and data-center demand**	Fabricated-metal shipments $471.5B (2023) to $490.4B (2025) [src-20261002-dedd62]; about 247,000 reshoring/FDI jobs announced in 2025, but only 14 of 64 tariff-linked announcements had converted by mid-2026 (med) [src-20261023-986299]	Partly cyclical; pledges run ahead of orders
5	**Cheaper robots and better software**	$18.6B robot installs globally, 38,428 in the US [src-20260924-28b59b]; 100% bonus depreciation [src-20260114-2075f7]	Tech structural; tax reversible
6	**Additive gets real**	3D-printing market $21.9B in 2025, +9.1% [src-20250325-4c256c]; one Navy boat has 100+ printed critical parts [src-20250319-f21bec]	Structural; Asia-led growth

The defense sub-tier, in numbers. DoD relies on more than 200,000 suppliers and has country-of-origin data for fewer than 10% of F-35 sub-tier suppliers; F-35 production stopped for about six months in 2023 and 2024 over Chinese magnet content; DoD put about $6.5B into 828 industrial-base projects in FY20–24 [src-20250724-1c09ad]. **Claim (Power):** 60-80% of a prime's output comes from small suppliers facing about 300-day payment terms [src-20260908-65fab6]. Sub-tier shops run batches "of 10 or 20" and ask "will you keep buying them from me" (NDIA president) [src-20260901-a44eee]. The bottleneck is thousands of untracked sub-tier shops, which is where a new owner-operator enters [inference].

4.3 Headwinds, at equal weight

1. **Spare capacity exists** (34% utilization), so primes can push work to existing shops [src-20250120-419768].
2. **Imports are cheaper:** Taiwan 20–30%, Vietnam/Malaysia more [src-20250709-43486e].
3. **Qualification is slow** (9.1).
4. **Tariffs raise input costs** (3.3).
5. **Rates and construction** (2.5).
6. **Automation saves less than it sounds. Claim:** labor falls from 100 to 30–50 hours a part, but net saving after machine costs is 20–30% at best [src-20250712-e5e110]; robot-learning success in real tasks is 59–66% [src-20260904-267293].
7. **Machine tools are imported.** US machine-tool shipments fell from $5B (1981) to $2B (1983) and imports reached about 60% by the late 1990s [src-20240118-4e5ac1], so even AI-native factories buy foreign machines [src-20260310-f52a8d].

4.4 Why an AI-enabled, vertically integrated company is the answer here

- **Versus job shops:** they lack capital, software and successors (2.1). An owner-operator that automates quoting, CAM and inspection runs machines more hours with fewer machinists [inference].
- **Versus marketplaces:** owned factories earn more; Proto Labs makes 49% gross margin in its own factories versus 31% in its network [src-20260206-8bb619][src-20260206-2b13c7].
- **Versus software sold to shops:** shops buy slowly and the vendor never owns the margin [inference].
- **Fabrication vs precision:** fabrication wins on speed and price for many small orders (**Claim, SendCutSend:** 48-hour delivery [src-20260227-e92d45]); precision wins on qualification and long contracts [inference].
- **Where it breaks [inference].** If primes keep using existing shops, or qualification stays slow, a capital-heavy factory sits idle.

4.5 What would have to be true, and signals to watch

- Primes sign multi-year contracts with new suppliers. **Watch:** more Lockheed-style deals; Sec. 867 commitments.
- Plants run well above 34% utilization. **Watch:** disclosed utilization or gross margin from Hadrian, Isembard, Senra.
- Qualification gets faster. **Watch:** first articles passed per quarter; Navy approvals of printed parts.
- Debt, not equity, funds plants.

4.6 Macro in more depth

The munitions gap. Claim (Power): the US buys about 100 Tomahawks a year, versus about 850 fired in one day in the Iran conflict [src-20260908-65fab6]. Lockheed has $60B+ of deals to triple PAC-3 output to 2,000 a year (trade press) [src-20261015-86f700].

How a buyer buys a part. A new producer wins one part number at one buyer at a time: 1. **RFQ:** the buyer sends a drawing and asks for price and date. 2. **Qualification:** AS9100 quality system, NADCAP for special processes (9.1 for timelines). 3. **First-article inspection (AS9102):** every dimension checked; repeated on any change [src-20190801-e81d7b]. 4. **Prototype and low-rate production,** often under OTAs that can roll into production without new competition [src-20241219-9f8fcc]. 5. **Rate production:** price and consistency decide; at about 1,000 parts a traditional shop is cheaper than Hadrian [src-20260908-ed94f0]. 6. **Repeat and multiyear orders:** the prize. Programs last about 50 years, and suppliers for a 2035 aircraft are being chosen now [src-20260818-9b2d5c].

Buyers also cap any one supplier (decision summary 5). The test is whether prototype wins become repeat rate orders; evidence is thin (section 9) [inference].

4.7 How the industry and its supply chain work

- **Metal Supply:** service centers buy mill metal, cut it and resell it at about 18-32% gross margin (section 8) [src-20260729-407fd7][src-20261021-16bb64]; about $200B of revenue across about 9,400 sites [src-20251201-4b732d] (med).
- **Fabrication:** SendCutSend, Senra, Machina, 1872; Atomic for molded plastics. Instant quotes and nesting (packing parts onto one sheet). Works for small batches, not mass production: Machina takes 30 minutes to hours per part; a stamping press takes seconds [src-20261023-c3b62f]. Structural and architectural metal: 14,533 sites, $28.26B payroll [src-20250626-23f6d9].

- **Precision & Defense Hardware:** CNC machining, metal 3D printing (AM), castings, composites. Hadrian, Divergent, Isembard, Freeform, Amca; candidates Fabri, Layup, Hybron. Incumbents Howmet, PCC, ATI, Carpenter, Proto Labs, Xometry (margins in section 8). **Claim (Rolls-Royce engineer):** small shops aim for low double-digit margins [src-20251120-0c271b]; **Claim (ex-Hadrian programming head):** a shop can go from 15% to 30% margin in 12-18 months with Hadrian-style operations [src-20261004-9b47f5]. Precision parts about $71B by 2026 (med) [src-20251024-2e5383].

4.8 Where AI and automation help

Instant quoting, automated CAM programming, scheduling, in-process inspection and robot welding/forming (data-loop table below; general case in 2.13).

4.9 Right to win, unit economics and data loop

Right to win: section 8.1. **Unit economics:** no startup discloses gross margin [gap]; first plants cost “well north of $10 million” [src-20260817-8867f2].

Company	Loop score (0–3)	How it works	Evidence / missing
Freeform	2	Cameras on the molten metal adjust each layer live	[src-20241024-7f1c2e][src-20260219-2125d7]. Missing: yield trend
Hadrian	2	Opus for workflow, programming and quality	30-40 day training [src-20260926-70087f]; 50% cut in order-to-first-part time in about two years [src-20261004-9b47f5]. Relies on outside CAM
Divergent	2	Design software feeds printer, then assembly	600+ part numbers [src-20250915-311606]. Missing: cost trend
Senra	1	Amp standardizes inputs and guides technicians [src-20260715-68183e]	99% FPY claim [src-20260701-027c1b]
SendCutSend	1	Instant quoting at volume	32,000 orders a month [src-20260525-4a12ab]
Isembard	1	MasonOS across franchises	Claim: 40-60% more output [src-20250929-857f17]
Machina	1	Robot forming learns each shape	Claim: 50x lead time [src-20261022-ca4c7a]
Nox	1	Auto-nesting and quote estimates	[src-20260325-5b68db]
1872	0	Welding AI belongs to Path, not 1872	[src-20260722-3bc2d8]

Metal Supply

> **Our view: Metal Supply** *(analyst judgment)* — **Watch, not invest yet. Conviction: low. Why:** tariffs and policy help US metal (Nucor credits trade policy for record shipments [src-20260727-787ce9]), but the margin sits with incumbents. Reliance is taking share without AI [src-20261022-1e6872], and we passed on Nox. **Counter:** distributor software is open territory (13), and a distributor that owns inventory and pricing data could build a real loop. **Change our mind:** an AI-native distributor shows better margins or turns than Reliance at scale. **Benchmarks:** Vulcan and Mariana upstream; Nox (fully integrated,

passed). **Good fit:** asset-light, sells to job shops and primes, wins on quoting and inventory, not on owning the metal.

Nox Metals (metal distribution; fully integrated; we passed)

- An automated metal distributor outside Detroit with nesting software [src-20261009-404b9b]. $11.5M seed (Hyperion) [src-20260616-1cf7cb]. About 12 people at one site [src-20260325-5b68db]. Its founder's "controversial belief" is "raw capacity over everything else" [src-20260427-83b102].
- Instant quoting is easy with a 50% buffer; the hard part is beating a human estimator (**Claim**) [src-20261022-3135b5]. Traction: 4.14.

Sub-type: metal supply / service center.

Fabrication

Our view: Fabrication *(analyst judgment)* — **Attractive now. Conviction: medium. Why:** it is the cheapest lane to enter (13, conclusion 4); SendCutSend shows the model can be profitable [src-20260519-38c7ed]; high use of machines comes from scheduling software rather than robots [src-20260626-541ac8]. **Counter:** customers "outgrow us" and buy their own machines [src-20251002-6a5afd], and SendCutSend can beat China on only about 20–25% of its work [src-20260519-2d6ceb]. **Change our mind:** repeat-order data showing customers staying as they grow. **Best placed:** Senra (pick), 1872 (fully integrated), Machina (hybrid); benchmark SendCutSend. **Good fit:** Series B, high-mix work, defense or hardware buyers, software-driven quoting.

SendCutSend (fabrication; fully integrated; fits, benchmark)

Sub-type: fabrication.

- **What it makes:** laser-cut sheet metal, CNC parts and finishing, ordered online with an instant quote. Owns all its plants (Reno; Paris, KY; Arlington, TX) [src-20261020-487a39].
- **Money:** $110M at a $1.01B valuation, May 2026, its first outside round [src-20260519-38c7ed]. Series B at $2.1B led by Spark, October 2026, size undisclosed [src-20261020-487a39] (med, one trade write-up).

1872 (fabrication; fully integrated; fits)

Sub-type: fabrication.

- **What it makes:** steel skids, frames and enclosures for energy and utilities, using AI and robots, in Cincinnati [src-20261022-fdbf29].
- **Money:** $15M seed via Ohio's O.H.I.O. Fund [src-20260722-985690].

Senra Systems (wire harnesses; fully integrated; fits)

Sub-type: fabrication (electrical assembly; not metal, closest fit).

- **What it makes:** wire harnesses and cable assemblies for defense and space, in its own plants (about 95,000 sq ft, two sites) [src-20260701-027c1b].
- **Money:** $65M Series B, July 2026 [src-20260715-68183e].

Machina Labs (robotic sheet forming; hybrid; partial fit)

Sub-type: fabrication.

- **What it makes:** robots that shape sheet metal without molds (“RoboCraftsman”), plus machining and welding. Runs its own plants and places cells with customers [src-20260204-ce7053].
- **Money:** $124M Series C, February 2026 [src-20260204-ce7053].

Precision & Defense Hardware

> **Our view: Precision & Defense Hardware** *(analyst judgment)* — **Selective; mostly priced out at the top. Conviction: medium. Why:** defense demand and long programs lock in suppliers for decades [src-20260818-9b2d5c]; Freeform’s continuous process scores 2 on the data loop; but Hadrian is valued at $7.87B [src-20260807-b0dc5b]. **Counter:** quality on complex parts is unproven (“none of their parts right now are flying”) and buyers cap any one supplier at about a third [src-20250708-342eaf]. **Change our mind:** parts flying on a production program at a lower unit price. **Best placed:** Freeform (fully integrated, pick), Isembard (hybrid pick); benchmarks Hadrian, Divergent; watch Amca, Atomic. **Good fit:** Series B, enters through spare parts, qualified with one prime.

Hadrian (precision parts; hybrid; fits mostly, benchmark)

Sub-type: precision.

- **What it makes:** machined parts in automated CNC factories run on its “Opus” software. Now also runs cells inside Lockheed’s plant [src-20261014-015327].
- **Money:** Series C $260M (equity plus debt), July 2025 [src-20250721-d99a2a]; about $1.6B valuation in January 2026, then $1.37B Series D at $7.87B in August 2026 [src-20260807-b0dc5b]; a $360M bank credit line [src-20260817-c48f28]. About $2.24B raised in total [Harmonic].
- **v9.9, from Odd Lots (company claims, CEO Chris Power):** Hadrian pitches itself as a “design agnostic” contract manufacturer making parts up to “30% of a Virginia-class submarine”; most factory work is skilled people “driving an incredibly dumb computer … 90% of advanced manufacturing”; even modelling his factory at zero labour and energy cost, some Chinese commercial parts land at about a third of his cost (transcript wording garbled) because of subsidised energy, capital and export rebates. On make-vs-buy, he says every company should keep 10–15% in-house, while SpaceX now sends parts out and Hadrian is preparing to make Raptor parts [src-20260921-a51ab2, 9:30, 10:42, 17:34, 32:37].

Isembard (precision parts; hybrid franchise; partial fit)

Sub-type: precision.

- **What it makes:** machined parts in owned and franchised factories, all on its “MasonOS” software; US, UK, Germany, France, Ukraine [src-20260310-45bab3].
- **Money:** $9M seed plus $50M Series A led by USV, March 2026; valuation undisclosed [src-20260310-45bab3]. Harmonic estimates about $250M (low).

Freeform (metal 3D printing; fully integrated; fits)

Sub-type: precision.

- **What it makes:** metal 3D-printed parts using AI-controlled multi-laser printers. It sells parts, not printers [src-20241024-7f1c2e].

- **Money:** $67M Series B, February 2026; PitchBook puts it at $179M after the round [src-20260219-2125d7] (med); about $126M raised.

Divergent (printed structures and munition airframes; fully integrated; fits, benchmark)

Sub-type: precision & defense hardware. Website: divergent.us (the old divergent3d.com no longer works).

What it does. Divergent makes metal structures for missiles, aircraft and cars with its own system, DAPS: software-defined design, its own metal 3D printers, and robotic assembly without fixed tooling [src-20260617-9683d4]. Its new printer, Monolith One (24 kW, 12 lasers), is built in-house and is not sold or licensed to anyone [src-20260617-9683d4]. It sells parts per unit with a low start fee ("factory-as-a-service") and does not license its technology [src-20260222-9c80a9]. **Label: fully integrated:** it owns the design software, the printers, the alloys, the factory and the assembly.

Defense traction. - **Tomahawk:** contracted by Raytheon as a second source for the Tomahawk midbody; first prototypes are done and production is aimed for the first half of 2027. The Raytheon–Pentagon contract still waits on the FY2027 budget [src-20260601-976843]. - **L3Harris:** teaming agreement (Oct 13, 2026) to use Divergent's printing for the Red Wolf and Green Wolf munitions [src-20261013-eda149]. - **Customers named:** Lockheed Martin, RTX, General Atomics, CoAspire, Saab, Triumph, Bugatti, McLaren [src-20260617-9683d4]; CoAspire work is for the Air Force's affordable-mass-missile programs [src-20260601-976843]. **Claim, CEO:** contracts with 15+ primes [src-20260222-9c80a9].

Factory 2 (Long Beach). 430,000 sq ft; 64 Monolith One printers over 24 months, on top of 6 in Torrance [src-20260617-9683d4]. **Claims, company:** up to 30,000 missile airframes or 60,000 warhead casings a year, 8x more output, about 2x the throughput of other printers [src-20260617-9683d4][src-20260601-976843]. About 80% of the factory is driven by defense volume (CEO) [src-20260601-976843].

Money and growth. $290M Series E (Sep 2025) at a $2.3B valuation; over $1B raised [src-20260222-9c80a9]. **Claims, CEO:** 2025 revenue more than 5x 2024; planned mix about 70% aerospace and defense, 30% auto over three years; unit cost at parity with high-rate casting [src-20260222-9c80a9]. About 500 staff [src-20260222-9c80a9].

Bear case. - **Qualification:** aerospace parts need certification that takes years. An independent view (USC's Prof. S.K. Gupta): "Certification is a long, complicated process. That's why technology adoption has been slow"; the article adds that printing is sensitive to conditions, which could make copying the factory harder [src-20260222-9c80a9]. - **Defense budget dependence:** Tomahawk volume waits on the FY2027 budget, and Factory 2 is mostly sized for munitions [src-20260601-976843]. - **Capital intensity:** in-house printers plus a 430,000 sq ft plant are expensive to build and fill [inference]. - **Cost vs casting at scale:** parity with casting is only the CEO's claim; no independent cost data found [gap].

Next step: confirm the Tomahawk production start and Factory 2 printer installs in 2027.

Amca (buys existing shops; fully integrated; fits)

Sub-type: precision.

- **What it does:** bought 8 certified aerospace plants and redesigns parts with its AI platform [src-20260520-af3e6c][src-20261008-149196].
- **Money:** $300M Series B at over $1B (Caffeinated, Lightspeed) [src-20260520-af3e6c].

Atomic Industries (tooling and molded parts; fully integrated; fits)

Sub-type: precision (tooling and molding).

- **What it does, in plain English.** Almost every plastic part (a car bumper clip, a drone shell, a subway seat bracket) is made by squirting molten plastic into a steel mold. The mold is the expensive, slow part: a skilled toolmaker designs where the cooling channels, ejector pins and plastic entry points go, then machines the steel, then runs trial shots and tweaks it, often over many weeks. Atomic uses physics simulation plus machine learning to design that mold, builds it in its own Michigan shop, and then runs the mold in its own presses to sell finished parts [src-20260130-9500f0][src-20261027-b51989].
- **Where it sits.** Detroit area (Warren and Shelby Township, Macomb County, Michigan). Founded 2019 by Aaron Slodov (software, ex-Google per TBPN), Austin Bishop and Lou Young Jr. (a veteran toolmaker, head of manufacturing); Y Combinator W21 [src-20231204-652d42][src-20260128-2b7e8b][src-20250919-2805c7].

Where the sub-sections blur (checked against company websites, Oct 27, 2026). Nox sells standard metal cut to size, with no machining or finishing; SendCutSend makes finished custom parts from the customer's own design [src-20261027-93d80b][src-20261027-82f772]. SendCutSend now also offers CNC, welding and finishing but stays in Fabrication because sheet cutting is its core. Senra makes wire harnesses, not metal parts (closest fit: assembly to a drawing) [src-20261027-6834f5]. Atomic says "today our focus is plastics" (**Claim**) [src-20261027-de288e]. Machina forms sheet to aerospace tolerances. Divergent spans both but most new volume is munitions [src-20260601-976843]. Amca buys single-source defense component makers (**Claim**) [src-20261027-77bb69].

Other candidates and warnings. Fabri (automated casting, $13.5M from RTX and Lockheed arms) [src-20260617-4dfb96]; Layup (composites, $42M A) [src-20260602-a7b79e]; Hybron (composite shells, $25M seed) [src-20260413-abc797]; Rangeview (castings from printed molds) [src-20260923-3ab808]; Furientis (low-cost interceptors, $25M seed) [src-20261006-410f77]; Helicon (composite drone frames; "less than ten customers", **Claim**) [src-20261020-c4b077]; Westmag (drone motors) [src-20260602-66bb0d]; CircuitHub (says it is profitable) [src-20260520-09ca24]; Jeevy (90% of volume to partner shops) [src-20261014-40926d]; Re:Build (roll-up, 12 sites) [src-20261006-1212f9]. Anduril's $3.7B shipyard makes it both buyer and competitor [src-20261017-5be775]. Warnings: NGP said Daedalus was doubling or tripling revenue months before insolvency [src-20251203-fc28ea]; "3D printing is most definitely way high on the hype curve" (drone founder) [src-20261022-f423ef]; contract manufacturing has "extremely low gross margin" (Lior Susan) [src-20240508-097e37].

4.11 Bear case, at equal weight

- Spare capacity, cheaper imports and slow qualification (4.3).
- Automation net savings 20–30% at best (**Claim**) [src-20250712-e5e110].
- Hype pricing: SendCutSend at $2.1B five months after $1.01B [src-20261020-487a39] (med).
- Failures: Daedalus, Markforged (section 8).
- **Metal supply:** incumbents are winning (Reliance share gains) [src-20261022-1e6872]; the moat is inventory, credit and relationships.

4.12 Key sources

Census shop counts [src-20250626-23f6d9][src-20250909-29c826]; utilization [src-20250120-419768]; shipments [src-20261002-dedd62]; BLS [src-20260901-d504f9][src-20260901-3b0073]; DoD industrial base [src-20250724-1c09ad]; Lockheed–Hadrian [src-20261014-015327]; NDAA [src-20251218-f0d6ef]; service centers [src-20251201-4b732d]; FAI [src-20190801-e81d7b]; company website checks [src-

20261027-93d80b][src-20261027-82f772][src-20261027-6834f5][src-20261027-de288e][src-20261027-77bb69].

4.13 Listen & watch

Full list: Appendix D.

Metal Supply

1. ★ **“Just Gonna Send It Podcast: Zane Hengsperger (Nox Metals) – Ep. 13”**, SendCutSend, Mar 25, 2026, 54 min. ▶ Open link — *Why:* Nox’s founder in conversation with a big customer type; why software beats old-line distributors on quoting and cutting. Repo: src-20260325-5b68db.
2. **“Reindustrializing America: Zane Hengsperger on building Nox”**, *First Builders*, Apr 27, 2026, length not listed. ▶ Open link — *Why:* the detailed founder story and lean-capital bet. Repo: src-20260427-83b102.
3. **“Building NOX Metals & The Future of Manufacturing” (#430)**, *Powers Pod*, length not listed. ▶ Open link — *Why:* Nox as the AI-run metal service center. Repo: src-20261022-3135b5.

Fabrication

1. ★ **“How to Build a Manufacturing Company in America | Jim Belosic, SendCutSend”**, *The Peel*, Apr 30, 2026, 1 h 32 min. ▶ Open link — *Why:* a bootstrapped fabricator at about $140M run rate in eight years (company figure); the best repeat-order case study. Repo: src-20260430-9318cd.
2. **“What Everyone is Missing About Reindustrialization”**, Machina Labs, Oct 20, 2026, 55 min. ▶ Open link — *Why:* Machina’s and Freeform’s founders on why the money is in the finished part ($57 of steel becomes a $2,000 part). Repo: src-20261020-bac9ca.
3. **“The Crisis Facing American Manufacturing”**, *Core Memory* (Ashlee Vance), Sep 26, 2026, 25 min. ▶ Open link — *Why:* a short documentary inside Hadrian and Machina Labs. Repo: src-20260926-70087f.

Precision & Defense Hardware

1. ★ **“How to Dominate Manufacturing in the US | Chris Power, Hadrian”**, *Relentless*, Sep 21, 2026, 1 h 19 min. ▶ Open link — *Why:* the clearest unit-economics case for owning factories; about 80% of capex is shared across programs. Repo: src-20260921-876080.
2. **“From hypercars to cruise missiles: Lukas Czinger on the future of US defense” (E2292)**, *This Week in Startups*, May 23, 2026, 1 h 46 min. ▶ Open link — *Why:* Divergent’s CEO on moving from cars to missile airframes and why it builds its own printers. Repo: src-20260523-b05ded.
3. **“Building the Arsenal: Speed, Scale, and Affordability”**, Air & Space Forces Association, Oct 14, 2026, 40 min. ▶ Open link — *Why:* the buyer side; Divergent’s CEO on a panel about producing munitions at scale. Repo: src-20261014-47f64b.
4. *(Read or listen)* **TBPN transcript: Atomic Industries raises $25M Series A**, Sep 19, 2025. ▶ Open link — *Why:* Atomic’s move from software to molds to parts; “mid eight figure” contracts (company claim). Repo: src-20250919-2805c7.

4.14 Commercial traction and incumbents

Labels as in 3.14.

Who’s actually commercializing

Company	Sub-section	Status	Evidence
Nox Metals	Metal Supply	Revenue	**Claim:** 46 people, 250–300 customers, 75,000 sq ft [src-20261009-404b9b]; a price quote in minutes versus 3–5 days at brokers [src-20250813-5ee11b].
SendCutSend	Fabrication	Revenue	About $140M a year (low confidence) [src-20260504-c3ded5]; 32,000 orders a month, growing ~100% a year [src-20260525-4a12ab]; 300,000+ customers [src-20261020-9deafa]; named customers include Zipline, Waymo, Anduril [src-20261020-c541ad].
Senra	Fabrication	Revenue	Growth comes from existing customers [src-20260701-027c1b].
Machina Labs	Fabrication	Revenue (paid projects)	Won’t start a project unless the customer commits dollars [src-20260622-7846f9].
1872	Fabrication	Lab / pre-revenue (no evidence)	$15M seed [src-20260722-985690]; no customer evidence found [gap].
Hadrian	Precision & Defense	Revenue	$3M revenue in 2023 [src-20240815-500933]; **Claim:** “10xed revenue” [src-20250718-ac643a]; Army Red River Depot contract to Mar 2027 [src-20260318-472c29]; cells inside Lockheed’s plant [src-20261014-015327].
Divergent	Precision & Defense	Revenue; defense ramp is pilot	Sells parts per unit [src-20260222-9c80a9]; Tomahawk midbody second source, production aimed for H1 2027 [src-20260601-976843]; L3Harris teaming (Oct 2026) [src-20261013-eda149].
Isembard	Precision & Defense	Revenue	**Claim:** 14 factories (4 owned, 10 franchised) on long-term contracts [src-20260909-b547c7].
Amca	Precision & Defense	Revenue	Eight plants; bought Electrocube [src-20260212-c61f30]; **Claim:** Army Materiel Command contract [src-20261008-9941fd].
Atomic Industries	Precision & Defense	Revenue (claimed)	**Claim:** “mid eight figure” part contracts [src-20250919-2805c7].
Freeform	Precision & Defense	Pilot	Hundreds of parts a week, customers undisclosed [src-20260219-2125d7]; DIU OTA [src-20260901-6e197d].

Revenue vs lab. This is the most commercial industry in the primer: Metal Supply and Fabrication are revenue businesses today. Precision & Defense has revenue, but most defense volume is still prototypes and second-source qualification. Failures show the risk: Markforged and Velo3D (section 8), Daedalus insolvency [src-20251203-fc28ea].

Incumbents. Precision Castparts and Howmet (castings and forgings) [src-20260923-3ab808]; Protolabs ($533.1M revenue, 44.5% gross margin) [src-20260206-8bb619]; Reliance and Ryerson (metal service centers) [src-20261022-1e6872][src-20260729-407fd7]. *What they do better:* qualified parts at volume. A buyer says for “a standard part on a 737” a Howmet or tier-two shop is more reliable [src-20260908-ed94f0]; Reliance is gaining share in metal supply [src-20261022-1e6872]. *Where startups win:* fast quoting and low-volume, high-mix work, and new defense programs where primes need second sources (primes earn 7–12% cost-plus, so they outsource) [src-20260501-4e5b9a].

Regulation and qualification. AS9100 certification typically takes 6–12 months from kickoff to certificate for small and mid-size shops (low; consultant source) [src-20260101-f25186]. ITAR registration and AS9100 are table stakes for defense work [src-20250929-857f17][src-20260819-dadd06][src-20261023-58b5a8]. Part qualification at a prime then adds its own cycle; Divergent’s Tomahawk work runs from prototype in 2026 to aimed production in 2027 [src-20260601-976843].

5-year outlook (to 2031). Likely. SendCutSend, Hadrian and Divergent already sell, and defense demand is pulling. **Bear case, equal weight:** defense demand is lumpy and budget-dependent (2.6); customers “outgrow” online fabricators by buying machines [src-20251002-6a5afd]; and incumbents can buy the same robots (GrayMatter rents cells to them [src-20261022-3335bd]), so the margin edge may shrink.

5. Specialty Chemicals

> **Our view: Specialty Chemicals** *(analyst judgment, not fact [inference])* - **Stance:** Selective. Pharma ingredients and fragrances now; agrochemicals and discovery labs to watch. - **Conviction:** medium. Margins are real (5.2) and reshoring policy helps, but new molecules face long reviews. - **Why:** (1) specialty players earn far more than commodity players (Lonza 31.6% vs losses at Dow) [src-20260128-af3b8f]; (2) China and India dependence gives US makers a policy-backed buyer (5.6); (3) Osmo shows that owning an AI-designed molecule and the plant can earn money [src-20260204-1ec8c6]. - **Strongest counter:** generics are excluded from the tariff, US demand is flat, and new-chemical reviews took a median of 520 days in 2025 [src-20261027-f4abbf]. - **What would change our mind:** Phlow’s FDA approval in 2027; Osmo naming large fragrance-house customers; faster EPA reviews. - **Best placed / watch:** Solugen, Osmo, Phlow (fully integrated); Scimplify, Enko (hybrid); Albert (enabler). - **Good SPC fit:** Series A–B; owns the molecule or process and a small US plant; buyer is a pharma, consumer or industrial brand with repeat orders.

What this industry is. Specialty chemicals are molecules sold for what they do, not what they are, in smaller volumes and at higher prices than bulk chemicals. This chapter covers pharmaceutical ingredients (APIs, their intermediates and key starting materials), agrochemical actives, electronic and semiconductor chemicals, flavors and fragrances, coatings and additives, catalysts, performance polymers, and AI molecule discovery. Engineered materials (battery cathodes, cooling fluids, alloys) and the materials discovery labs sit in chapter 6. **Brooke’s focus** runs through the chapter: the chemical supply chain (5.6, 5.7) and AI-designed molecules (5.8, 5.10, 5.14).

5.1 The problem today

- **US drug supply rests on China and India.** About 46% of US generic doses depend on Chinese source materials (detail 5.6).
- **Specialty demand is soft.** US specialty output is expected to fall 0.3% in 2026, except for chips, data centers and healthcare (detail 5.6).
- **New molecules take years to reach customers.** Discovery is getting faster, but qualification and regulatory review (TSCA, FDA, pesticide registration) are not (5.14).
- **The ‘valley of death’ killed the last wave.** See 5.11 (1).

5.2 Market map, size and economics

> **Our view by segment** *(analyst judgment)*: **Pharma ingredients:** attractive now, medium conviction (policy buyer; Phlow waits on FDA). **Fragrances and custom specialty:** attractive now, medium (Osmo and Scimplify already sell). **Agrochemical actives:** watch, low (registration takes 10–15 years, a **claim** by Enko’s CEO [src-20250601-f7467b]). **Electronic chemicals and catalysts:** watch, low (lab stage only). **Coatings, additives and polymers:** no producer startup found; software only (Albert).

Estimates of the global specialty market vary with definitions: **$626.5B** in 2025 on one count [src-20260101-69df28] and **$1.4T** on a broader one [src-20250101-1d84c3] (both low-confidence market-research figures). Asia-Pacific is about 47% of revenue [src-20260101-cab1ab].

Segment	Rough share of specialty market	What it is	Startups in this primer
Pharma APIs, intermediates, KSMs	~8–10% [src-20250101-1d84c3]	The active molecule in a drug and its building blocks	Phlow, On Demand Pharmaceuticals, Warp, Chemify, Scimplify
Agrochemical actives	~10–20% [src-20250101-1d84c3][src-20260101-cab1ab]	Herbicides, fungicides, insecticides	Enko
Electronic / semiconductor chemicals	~15% [src-20260101-cab1ab]	High-purity acids, CMP slurries, photoresists	Scimplify (partly); Discovered Materials (discovery)
Flavors and fragrances	not broken out [gap]	Scent and taste molecules	Osmo
Coatings, adhesives, additives	~25–30% coatings, ~18–22% adhesives [src-20250101-1d84c3]	Resins, pigments, curing agents, additives	**No producer startup found**; Albert (formulation software)
Catalysts	not broken out [gap]	Molecules that speed reactions in plants and cars	Dunia (discovery); Orbital licenses catalyst software to BASF (6.10)
Performance polymers	~33% of one estimate (specialty polymers) [src-20260101-cab1ab]	High-spec plastics and elastomers	**No startup found**
Personal care and bio-based ingredients	~12–15% [src-20250101-1d84c3]	Emulsifiers, actives, bio-made ingredients	Debut, Solugen

Gaps: the segment shares come from different vendor reports with different scopes, so they do not add to 100%.

How specialty economics differ from commodity

	Commodity	Specialty
Sold on	Price per ton	Performance in the customer’s product

	Commodity	Specialty
Plants	Giant, continuous (millions of t/yr)	Small, often batch, multi-product (5.7)
Margins (2025)	Dow net loss $2.4B [src-20260129-338338]; LyondellBasell net loss $738M [src-20260130-12a9f3]	Lonza CORE EBITDA 31.6% [src-20260128-af3b8f]; Clariant 17.8% [src-20260226-f774af]; Evonik 13.3% [src-20260305-5c5e74]
Switching cost	Low	High: the customer must re-qualify. **Claim (Scimplify):** custom development usually takes 12–18 months [src-20261014-b218c5]
Where AI helps	Plant control	Finding the molecule, the route and the formulation, then speeding qualification

Meaning [inference]: specialty buyers pay for performance and pay again for reliability, so a supplier that is qualified tends to stay. That makes specialty the part of chemistry where a startup can earn margin without matching incumbents on scale.

5.3 Tailwinds, ranked [ranking is inference]

Rank	Tailwind	Date / size	Structural or reversible?
1	**Pharma onshoring policy**	Section 232 pharma tariff and BIOSECURE (detail 5.6)	BIOSECURE structural (statute); tariff reversible
2	**The state as a buyer of APIs**	Strategic API Reserve for about 26 critical drugs [src-20260701-818dbe]	Executive order; reversible
3	**Specialty demand from chips, data centers and healthcare**	Still growing while other specialty output falls [src-20260615-0c9b7e]	Structural while AI capex lasts
4	**Europe’s retreat**	Closures and insolvencies leave buyers of European intermediates looking for suppliers (5.6)	Structural (energy costs)
5	**AI shortens discovery and formulation**	**Claims:** Enko 75% faster discovery [src-20250601-f7467b]; Scimplify 2 months vs 12–18 [src-20261014-b218c5]	Unproven at scale
6	**Government grants and loans**	Phlow $15.9M award [src-20260721-fa5081]; On Demand Pharmaceuticals ~$35.6M [src-20250813-79a0cf]; Solugen conditional $213.6M DOE loan [src-20240613-2df5d6]	Annual budgets; reversible
7	**FDA PreCheck**	Reviews a new plant before any product filing; seven sites picked June 29, 2026 [src-20260629-8074e1]	Pilot program

Pharma onshoring money, in numbers. Fourteen big drugmakers have pledged more than $480B of US production [src-20260601-17c2d4], but only about $60B of about $600B of pledges has specific amounts and locations [src-20260210-883e2f]. *Meaning [inference]:* the pledges fund big pharma’s own patented-drug plants. The opening for startups is generic APIs and KSMs, where only policy (the reserve, a generics tariff, BIOSECURE) creates a buyer.

5.4 Headwinds, at equal weight

1. **Specialty demand is flat.** US specialty output is expected to fall 0.3% in 2026 [src-20260615-0c9b7e]; in Europe, dyes and pigments fell 2.7% in the first half of 2026 [src-20260915-e74614].
2. **Qualification is slow and sticky.** Incumbents are already qualified; a customer switches only for a real gain (5.2).
3. **Regulation sets the clock, not AI.** A new US chemical's median TSCA review took 520 days in 2025; pesticides and drugs take longer (5.14).
4. **Generics are outside the 232 tariff.** See 5.11 (4).
5. **Pledges are not plants.** See 5.3.
6. **Rates.** Plants are capital-heavy and long-dated; Fed 3.75–4.00% [src-20260916-f0c016].
7. **AI discovery is not the bottleneck.** Discovery without scale-up does not create supply (see 5.8).

5.5 Why an AI-enabled, vertically integrated company wins here

- **Owning the plant captures the margin.** Specialty margins are earned by whoever is qualified to supply (5.2); pure AI software for chemistry is a tool, labeled **enabler**.
- **AI shortens the slow steps.** Finding the molecule (Enko, Osmo), planning the route and process (Scimplify, Chemify), and continuous-flow plants (Phlow, On Demand Pharmaceuticals). All are **claims** at this stage.
- **Each order feeds the data loop.** A producer that makes many custom molecules builds proprietary process data that a software vendor never sees [inference].
- **Versus imports:** BIOSECURE and the 232 tariff reward provable US supply.
- **Where it breaks [inference]:** regulators (FDA, EPA) set the timeline, and a customer's qualification cycle does not move just because the molecule was found faster.

What would have to be true, and signals to watch. - Onshoring rules bite on generics. **Watch:** any 232 extension to generics; BIOSECURE's first OMB list (Dec 2026). - Repeat orders from paying buyers. **Watch:** Phlow's FDA decision (2027); Osmo's ingredient sales to large brands; Scimplify's US plant with Teknor Apex. - AI-designed molecules clear regulators. **Watch:** Enko herbicides entering registration.

5.6 Macro in more depth: the supply chain

China and India own the pharma building blocks. About 90% of US prescriptions are generics. India supplies 47.2% of US generic doses and China 13.0%; only 10.5% are US-made. Indian firms get 65–70% of their KSMs, intermediates and APIs from China, so after adjusting, about 46% of US generic doses depend on Chinese source materials [src-20250605-bde73f]. Estimates vary: direct Chinese API is about 8% of volume, and Brookings puts Chinese API in "perhaps a quarter" of US drug volume [src-20250501-c298cb]. CFR finds about 41% of KSMs in US-approved APIs come only from China, including the starting materials for 94% of amoxicillin, 74% of heparin and 70% of acetaminophen sold in the US; US amoxicillin production went from 100% domestic to zero in twelve years. Shortages would peak 4.5–9 months after a Chinese restriction [src-20260501-7f527d]. Only about 10% of APIs by volume are US-made [src-20260701-818dbe]. *Meaning [inference]:* the case is strongest for a short list of old, cheap molecules, which need a price floor or a government buyer.

Beyond pharma, Asia holds the volume. Asia-Pacific is about 47% of specialty revenue [src-20260101-cab1ab]. Electronic chemicals are growing with Asian fab expansion and US subsidies [src-20260101-cab1ab]. India is emerging as an AI-enabled specialty supplier as buyers diversify away from China (Scimplify, 5.10) [src-20261014-b218c5].

Europe is shrinking. Announced European closures reached 37 Mt over 2022–25, about 9% of capacity, with energy cost the reason in 49% of cases [src-20260115-ec35af]. Plants need 80–85% utilization to pay; Europe averages about 74%, and more than 30 German chemical companies a year have gone insolvent since 2024 [src-20260401-e2adb0]. Specialty exports from the EU fell €1.5B in the first half of 2026 [src-20260915-e74614]. *Meaning [inference]:* buyers of European intermediates need new suppliers, and US makers with cheap energy are the natural replacement.

US demand. US specialty output is expected down 0.3% in 2026 and up 2.7% in 2027; specialties for semiconductors, data centers and healthcare keep growing [src-20260615-0c9b7e].

Policy is pushing reshoring, mainly in pharma. - **Section 232 pharma tariff (April 2026):** 100% on patented drugs and their ingredients; 20% with approved onshoring plans (rising to 100% in 2030); zero for pricing-and-onshoring deals; 15% for the EU, Japan, Korea and Switzerland. Generics are excluded for now. Only 15% of patented API volume is US-made [src-20260402-ecae19]. Deadlines run to July 31 and September 29, 2026; pricing-deal firms get 0% until 2029 [src-20260601-17c2d4]. - **BIOSECURE Act:** enacted Dec 18, 2025. It bars federal contracts, grants and loans involving "biotechnology companies of concern"; the first OMB list is due by Dec 2026 [src-20260110-4db409]. The Pentagon added WuXi AppTec to its 1260H list in June 2026 [src-20260818-489fdf]. - **Strategic API Reserve:** an August 2025 order directs HHS to hold six months of API for about 26 critical drugs, favoring domestic sources [src-20260701-818dbe]. - **Executive Order 14293 and FDA PreCheck** speed approval of new US drug and API plants (section 11).

5.7 How the industry and its supply chain work

Think of chemistry as a five-step ladder. Each step takes the output of the step below and makes it more specific, smaller in volume and higher in price per kilogram.

1. **Feedstocks.** The raw inputs: natural gas and natural-gas liquids (ethane, propane), crude-oil fractions (naphtha), salt, sulfur, phosphate rock and, increasingly, biomass such as corn sugar.
2. **Commodity (basic) chemicals.** Made in giant, continuous plants measured in millions of tons a year. A **steam cracker** heats ethane or naphtha to "crack" it into ethylene and propylene, the building blocks of most plastics. Others: chlorine, sulfuric acid, methanol. Price is set by feedstock and energy cost; margins are thin and cyclical.
3. **Intermediates.** Molecules one or several reaction steps further on (for example acetic acid, adipic acid, solvents and monomers). In pharma, the key ones are **key starting materials (KSMs)**, the purchased building blocks an approved drug process starts from.
4. **Specialty and fine chemicals.** Smaller volumes sold for what they do, not what they are: coatings, catalysts, water-treatment chemicals, electronic chemicals, flavors, crop chemicals, energetics. Higher margins; buyers qualify a supplier and then rarely switch.
5. **Active pharmaceutical ingredients (APIs).** The molecule in a drug that does the work. A **finished dose form (FDF)** is the pill or injection. A **CRO** (contract research organization) makes molecules for research; a **CDMO** (contract development and manufacturing organization) makes them at commercial scale under FDA rules.

Two engineering terms matter for startups. **Batch** chemistry runs one tank at a time; **continuous flow** pumps reactants through pipes or small reactors nonstop, which is safer for hazardous steps and easier to automate (FDA guidance Q13 covers continuous manufacturing). **Biomanufacturing** uses enzymes or engineered microbes (fermentation) instead of heat and pressure to make molecules.

Who buys: ACC says more than 80% of basic and specialty chemicals are bought by other industries, not consumers [src-20260615-0c9b7e]. So chemical demand follows manufacturing, housing, autos, data centers and pharma.

Market size (this industry). No clean free revenue number for US specialty chemicals in the repo [gap]. Global specialty estimates: 5.2. Europe announced 37 Mt of closures in 2022–25 [src-20260115-ec35af]; only about 10% of APIs by volume are made in the US [src-20260701-818dbe].

5.8 Where AI and automation help

Stage	What AI does	Maturity	Molecule-to-revenue time
Discovery (what molecule?)	Generative design, property prediction, screening billions of candidates	Real but crowded; models are commoditizing (4A)	Drugs: 10+ years to market; specialty chemicals: 2–5 years to qualify
Synthesis planning (how to make it?)	Retrosynthesis software suggests reaction routes (Synthia, IBM RXN, PostEra)	Decision support only; chemists still filter routes [inference from literature review]	Months saved per route
Lab execution (actually make it)	Robots plus AI run reactions and learn from results (“self-driving labs”)	Early; claims of autonomous discovery have been disputed (the 2023 A-Lab row)	Faster make-test loops for research compounds
Process and plant (make it cheaply at scale)	New catalysts and enzymes, continuous-flow control, yield optimization	Where the money is, and where startups own plants	3–7 years for a new plant [inference]

The key point for SPC: **discovering a molecule is not the bottleneck; making it reliably at scale and getting buyers to qualify it is.** This matches the materials finding in 4A.4.

5.9 Right to win, unit economics and data loop

Right to win [inference]. Own the plant and the offtake; use automation to shorten scale-up; target molecules where policy forces US supply (APIs, intermediates, defense chemicals), not commodities where Dow and ExxonMobil win on cost.

Data loop (score 0–3; scale in 2.13):

No chemicals company in the repo has a scored data loop yet [gap].

5.10 Companies, grouped by label

Producers — fully integrated

Solugen (Houston; bio-based specialty chemicals; Full). *Qualifies as specialty, borderline: hydrogen peroxide is also a bulk chemical, but Solugen sells small-plant, bio-based product into specialty uses [inference].* - **Process:** “chemoenzymatic” production pairs an enzyme with a metal catalyst to turn corn syrup into chemicals, starting with hydrogen peroxide and organic acids. **Claim (Solugen founders):** 96% yield at scale versus about 60% for the traditional route [src-20260320-f90e80]. - **Origin:** started with a reactor of about $7,000 built from Home Depot PVC pipe, which at its peak earned $12,000 a month (founder’s account), and sold to hot-tub owners in Dallas [src-20260320-f90e80, 0:00, 7:06] before raising large rounds. Its Bioforge plant runs 24/7 on railcars of corn syrup. **Claim:** smaller plants near customers undercut incumbents on shipping [src-20260320-f90e80]. - **What it means [inference]:** the clearest AI-adjacent chemical producer. It owns plants, sells product and has repeat industrial buyers. But the edge is enzyme engineering and plant design more than AI, and some products face bulk-chemical pricing.

Phlow (Richmond/Petersburg, VA; US-made essential-medicine APIs; Full). *Strong fit; fuller profile.* - **What it makes:** active pharmaceutical ingredients (APIs) for essential generic medicines, in its own plants using batch plus continuous-flow chemistry: a kilo facility (~250 kg a year) and a metric-

ton facility (50–100 t a year) [src-20260209-16223c][src-20260721-fa5081]. - **Who buys / repeat orders:** a deal under which Phlow makes epinephrine API and Fresenius makes finished doses; validation campaign done; Drug Master File filed; commercial supply pending FDA in 2027 [src-20260209-16223c]. A $15.9M award to an end-to-end Virginia alliance: Occam makes building blocks, Phlow the APIs, Civica finishes ketamine, midazolam and others [src-20260721-fa5081]. Government buyers plus a generic maker = the kind of offtake that can repeat. - **AI / automation angle:** continuous-flow manufacturing (smaller footprint, more automation, fewer batch steps) — automation rather than AI. - **Why it fits:** the most direct answer to the 46% China-dependence number [src-20250605-bde73f] and to BIOSECURE. - **v9.9, CEO interview (company claims unless noted):** the COVID-era BARDA contract (about $899M, host's figure) paid for development and process-chemistry labs, two cGMP plants (batch and continuous, kilogram to large metric-tonne scale) and the first Strategic API Reserve; Phlow re-engineers 40–50-year-old generic routes with telescoping batch or flow chemistry and also serves pharma reshoring (for example a $1B drug made at a Chinese contract manufacturer under scrutiny). Macro: drug shortages at "an all time high"; India halted exports of 20-odd APIs in March 2020; the US has no beta-lactam (penicillin) drug-substance or cephalosporin fermentation base; he wants a CHIPS-Act-style programme [src-20240613-4b9301, 33:58–36:41, 37:25, 39:47–40:31, 41:37–43:31]. - **Bear (one line):** generics are excluded from the 232 tariff and margins are thin; one FDA delay pushes revenue a year. v9.9: revenue grew from one CARES Act contract that outlived the emergency, it competes with state-subsidised rivals ("we're actually competing against their governments"), and the CEO himself calls US support so far "a drop in the bucket" [src-20240613-4b9301, 34:23, 37:14, 38:49].

On Demand Pharmaceuticals (Rockville, MD; on-demand API and drug manufacturing; Full). - **What it makes:** APIs at its Rockville plant, plus a modular "Pharmacy on Demand" unit using AI/ML process control, deployed at North Mississippi Medical Center [src-20250813-79a0cf]. - **Who buys:** hospitals and government (DoD/HHS funded since 2020, about $35M in federal awards); the unit makes shortage drugs [src-20250813-79a0cf]. No commercial repeat-order data. - **AI / automation angle: Claim:** AI/ML-controlled modular continuous manufacturing. - **Bear (one line):** still grant-funded; hospital-scale units may not reach meaningful volume.

Osmo (New York and Elizabeth, NJ; AI-designed fragrance molecules and formulas; Full). *Added v9.6.* - **What it makes:** fragrance formulas and its own AI-designed fragrance molecules (Glossine, Fractaline, Quasarine), sold through its "Generation" ingredient arm since March 2025. It formulates, manufactures, bottles and packages [src-20260204-1ec8c6]. A 60,000 sq ft plant with robotics opened in Elizabeth, NJ in May 2026; filling and logistics use local partners [src-20260512-3ad5dc]. - **Who buys:** nearly 100 brands, mostly small ones (Kindred, Loucil and others); minimum orders of 500 units or 10 kg of oil [src-20260204-1ec8c6]. No revenue disclosed. - **Money:** $70M Series B led by Two Sigma Ventures in Feb 2026, $130M total [src-20260204-1ec8c6]. - **Why it fits [inference]:** the clearest case of AI-designed molecules sold as product by the company that designed them, made in its own US plant. - **Bear (one line):** customers so far are small brands; big fragrance houses (Givaudan, Firmenich-DSM, IFF, Symrise) own the large accounts, and a fragrance molecule's value depends on perfumers adopting it.

Producers — hybrid

Scimplify (Bengaluru; AI-run custom specialty chemical manufacturing; Hybrid). *Added v9.6.* - **What it does:** takes a customer's specification, develops the process in its own labs in Bengaluru and Hyderabad, then makes the product at partner factories. Its AI platform, Scinode, picks routes and plants and tracks quality and regulatory needs [src-20261014-b218c5]. **Claim:** more than 40% of its portfolio is advanced materials for semiconductors, batteries and critical minerals [src-20261015-2d7316]. - **Who buys:** Henkel and Teknor Apex are customers and invested; Teknor Apex will be its US manufacturing partner [src-20261014-b218c5][src-20261015-2d7316]. **Claim:** 2 months from

requirement to delivery versus the usual 12–18 [src-20261014-b218c5]. - **Money:** $25M led by Hitachi Ventures (Oct 2026), $80M total equity [src-20261014-b218c5]. - **Why it fits [inference]:** a direct answer to “China plus one” in specialty supply, and it owns the customer even though partners run the reactors. - **Bear (one line):** an Indian, asset-light model is a competitor to US reshoring rather than part of it, and partner factories limit control of quality and margin.

Enko (Mystic, CT; AI-discovered crop-protection actives; Hybrid, close to Enabler). *Added v9.6.* - **What it does:** discovers herbicides, fungicides and insecticides with its ENKOMPASS platform (machine learning plus DNA-encoded libraries) [src-20250601-f7467b]. It works with Syngenta, which took its new-mode-of-action herbicide leads to proof of concept [src-20241210-12825a]. - **Traction:** two advanced herbicide programs with 4+ years of field data across 200+ trials [src-20250601-f7467b]. **Claim (CEO):** discovery 75% faster and 90% cheaper [src-20250601-f7467b]. - **Bear (one line):** registration still takes years and the regulatory step is unchanged, as the CEO admits [src-20250601-f7467b]; the likely model is licensing to majors, not selling product.

Warp Sciences (San Francisco; US drug-discovery chemistry CRO; Hybrid). - **What it is:** a US-based chemistry CRO that combines expert chemists, AI and robotics in labs “designed for automation from day one” to make the molecules biotech and pharma teams test. It does not develop its own drugs [src-20261027-c11426][src-20261022-b542e2]. - **Team:** CEO Kevin Weil (ex-OpenAI Chief Product Officer and head of OpenAI for Science), COO DJ Patil (first US Chief Data Scientist), CSO Sol Reisberg (about 20 years in medicinal chemistry, ex-Gilead and Schrödinger, a long-time buyer from offshore CROs) and Mano Paluri (ex-Meta FAIR) [src-20261022-b542e2]. - **What it means [inference]:** a direct BIOSECURE and reshoring play with a top-tier team, but it is services-first and launched five days ago. No customers, prices or throughput are public. US chemist labor costs several times Chinese CRO rates, so the thesis depends on automation actually lowering unit cost. It is research-scale, not manufacturing-scale.

Chemify (Glasgow; robotic chemistry, “chemputation”; Hybrid). - **What it is:** founded in 2022 from Lee Cronin’s University of Glasgow lab. Its “Chemifarm” turns code written in a chemical programming language into molecules made by robots. The first £12M Chemifarm opened in June 2025 [src-20251021-d0aa43]. - **Money:** a Series B of more than $50M (Wing, Insight) [src-20251021-d0aa43]. In Sept 2026, £22M in Scottish and UK government grants went toward an £89.9M expansion, with staff growing from 152 to about 450 in three years [src-20260917-2991e2]. - **What it means [inference]:** the most ambitious “AI makes molecules” platform. Today it is a research-scale CRO and co-discovery shop, not a bulk producer, so it sits beside the discovery labs in 6.10.

Watch list and AI-chemistry enablers

Watch list (short notes, not core): - **Debut (beauty ingredients by biomanufacturing; Hybrid).** $20M for AI ingredient discovery; pilot plant; $2M to plan a facility [src-20250801-a4ae9d]. Bear: no commercial plant yet. - **Visolis:** not added; only a LinkedIn source was found.

AI-chemistry enablers (software or drug developers; not producers): - **Albert (Oakland; AI for formulation R&D; Enabler).** $55M Series B (Oct 2026); used by AkzoNobel, BASF, Chemours, Henkel, Kenvue, Nouryon and others [src-20261013-fbc067][src-20261015-2d7316]. The main AI name in coatings and additives, but software, not a producer. Bear: incumbents keep the margin. - **Dunia Innovations (Berlin; AI plus robotic labs for catalysts and battery materials; Enabler).** Serves customers in catalysts, batteries and semiconductors; €280M “GigaLab” due 2028 [src-20260520-57be93]. Bear: no named customers or product sales. - **Discovered Materials (YC; AI agents for semiconductor thermal materials; Enabler).** $9M seed (Aug 2026); released hundreds of candidate materials, no customers named [src-20260810-357833]. Bear: candidates are not products. - **Iambic (Enabler / drug developer).** Enchant v3 model (41B parameters) and an automated lab that makes hundreds of compounds a week; its own drug pipeline (IAM-C1, IAM217) and partners; S-1 filed

October 2026 [src-20260921-cb3d77]. Sells drugs, not chemicals. Bear: drug-development risk dominates. - **PostEra and SYNTHIA (Merck KGaA's retrosynthesis software; Enablers).** Software for planning how to make molecules [src-20261027-13f9b3]. Tools, not producers. Bear: commoditizing as general AI models improve [inference].

Process-plant software (cross-reference). Lumaril, AI closed-loop control sold to chemical, polymer and biologics plants and paid on the production it adds (company claim), is profiled in full in section 7.10 as an enabler [src-20261027-d5ba3f].

5.11 Bear case, at equal weight

1. **Biomanufacturing has a terrible record.** Ginkgo, Zymergen and Amyris all collapsed in 2021–23. For simple molecules, making them with organisms usually costs more than conventional chemical synthesis, so bio works mainly for complex, high-value compounds [src-20240101-499ec8]. Zymergen had no in-house commercial-scale manufacturing, its contract manufacturers missed production goals, and customers could not get its Hyaline film to work in their own production lines [src-20220101-9c58e3].
2. **AI-for-chemistry software alone fails.** Zapata is the example [src-20241011-721c93]. Retrosynthesis tools are still decision support for chemists [inference].
3. **Commodity economics punish newcomers.** Europe's plants are closing because of energy cost and Chinese overcapacity [src-20260115-ec35af]. A US startup selling a commodity competes with shale-fed incumbents and Chinese surplus, so it needs a real cost edge, not a green premium.
4. **Pharma policy can reverse or be diluted.** Section 232 exempts generics (where China dependence is deepest) and gives 0–20% rates to companies that do deals [src-20260402-ecae19]. A federal court enjoined WuXi's 1260H designation (search summary; low). The cheaper Indian and Chinese supply still exists.
5. **Capex before contracts again.** Chemical plants take years and hundreds of millions. Solugen's DOE loan is still conditional [src-20240613-2df5d6].
6. **Labor cost of US CROs.** Warp and Chemify must automate faster than offshore labs cut prices [inference].

What fits SPC [inference].

Company	Integration	Fit with "AI-enabled, vertically integrated, makes and sells product"
Solugen	Full	**Best fit.** Own plants, product revenue, government loan; AI is secondary to enzyme and process IP. Late stage.
Orbital (6.10)	Full (specialty fluids)	Good fit; see 6.10.
Warp Sciences	Hybrid (CRO)	Fits the reshoring and data-loop thesis; services, not product; too early to judge.
Chemify	Hybrid (CRO plus IP)	Interesting option on "code to molecule"; research scale.
CuspAI, Periodic, Lila	Enabler or discovery lab	Not producers; track via 6.10.

Diligence questions: What is the yield and cost per kilogram at the current plant versus incumbents? How many customers have reordered after qualification? Who keeps the data rights? How much capex is committed before offtake contracts?

5.12 Key sources

- Cefic gas and trends [src-20260915-e74614][src-20251001-7ec954]
- Cefic closures [src-20260115-ec35af]; Roland Berger [src-20260201-63910c]; FTI [src-20260401-e2adb0]
- ACC 2026 outlook [src-20260615-0c9b7e]
- USCC on generics [src-20250605-bde73f]; Brookings [src-20250501-c298cb]; CFR [src-20260501-7f527d]
- Section 232 pharma [src-20260402-ecae19]; BIOSECURE [src-20260110-4db409]; WuXi 1260H [src-20260818-489fdf]
- Solugen DOE loan [src-20240613-2df5d6]; Zymergen [src-20220101-9c58e3]; pledges and deadlines [src-20260601-17c2d4][src-20260701-818dbe]
- New company sources: Phlow, ODP, Debut, Iambic, PostEra/SYNTHIA (cited in 6.10)

5.13 Listen & watch

Full list: Appendix D.

1. ★ **“Inside The Startup Reinventing The $6 Trillion Chemical Manufacturing Industry”**, YC, Mar 2026, about 13 min. ▶ Open link — *Why:* the best short intro to a fully integrated chemical maker (Solugen; ~96% yield claim). Repo: src-20260320-f90e80.
2. **“EP 312 Lee Cronin on Automating Chemistry”**, *The Jim Rutt Show*, Jul 2025, length not listed. ▶ Open link — *Why:* Chemify’s “code to molecule” and how it charges (company claims). Repo: src-20250724-0f42f3.
3. **“Mission-driven doctor, inventor & entrepreneur” (Ep. 175)**, *Molecule to Market*, Phlow CEO Eric Edwards, 55 min. ▶ Open link — *Why:* drug-ingredient onshoring with the government as anchor buyer. Not in the repo.

5.14 Commercial traction and incumbents

Labels as in 3.14. This is the deepest traction section because specialty chemicals is where lab-to-revenue gaps have killed the most startups.

Who’s actually commercializing (selling product first)

Company	Segment	Status	Evidence
Solugen	Bio-based specialty chemicals	**Revenue**	Bioforge plant runs 24/7; repeat industrial buyers; conditional $213.6M DOE loan [src-20240613-2df5d6][src-20260320-f90e80].
Osmo	Flavors and fragrances	**Revenue**	Nearly 100 brands; own NJ plant since May 2026 [src-20260204-1ec8c6][src-20260512-3ad5dc]. Revenue not disclosed.
Scimplify	Custom specialty / electronic	**Revenue**	Henkel and Teknor Apex are customers [src-20261014-b218c5].
On Demand Pharmaceuticals	APIs	**Pilot**	About $35M of federal awards; one hospital-deployed unit [src-20250813-79a0cf].

Company	Segment	Status	Evidence
Phlow	APIs	**Pilot**	Epinephrine API with Fresenius; validation done; commercial supply pending FDA in 2027 [src-20260209-16223c]; $15.9M award [src-20260721-fa5081].
Chemify	Robotic chemistry (CRO)	**Pilot**	**Claim:** 12 customers [src-20250724-0f42f3]. Research scale.
Enko	Agrochemical actives	**Pilot** (partnered)	Syngenta collaboration; field trials, no product registered [src-20241210-12825a][src-20250601-f7467b].
Debut	Beauty ingredients	**Pilot**	Pilot plant [src-20250801-a4ae9d].
Warp Sciences	Chemistry CRO	**Lab / pre-revenue**	Launched October 2026; no customers public [src-20261022-b542e2].
Albert, Dunia, Discovered Materials	Enablers	Albert **revenue** (software); Dunia and Discovered **lab**	[src-20261013-fbc067][src-20260520-57be93][src-20260810-357833]

Failure to remember: Zymergen [src-20220101-9c58e3].

Revenue vs lab, by segment. *Real revenue:* bio-based specialty chemicals (Solugen), fragrances (Osmo), custom specialty manufacturing (Scimplify) and formulation software (Albert). *Pilot:* US-made APIs (revenue waits on FDA) and agrochemical actives (waits on registration). *Mostly lab:* catalysts and semiconductor-material discovery, and the new CROs. *No startup found:* coatings and additives producers, performance polymers.

Startup vs incumbent: scale and margins.

	Lonza	Evonik	Clariant	BASF	Startups
2025 sales	CHF 6.5B (+21.7%)	€14.1B (−7%)	CHF 3.9B (flat)	€59.7B	Osmo $130M raised in total; Phlow 50–100 t/yr API [src-20260721-fa5081]
Margin	CORE EBITDA 31.6%	Adj. EBITDA 13.3%	17.8%	11.0%	Not public
Investment	Capex CHF 1.37B	R&D €418M	—	R&D €2.0B	Solugen conditional DOE loan $213.6M
Source	[src-20260128-af3b8f]	[src-20260305-5c5e74]	[src-20260226-f774af]	[src-20260227-9fe9e6]	as cited

Albemarle’s specialties unit sold $1.37B in 2025 [src-20260201-955851]. Croda: no free source in the repo [gap].

What incumbents do better: qualified products, global sales forces and regulatory dossiers. Lonza, a contract maker for pharma, grew 21.7% while commodity-heavy names shrank, which shows that outsourced pharma manufacturing is where buyers pay [src-20260128-af3b8f]. Incumbents buy AI rather than build it: BASF licenses Orbital’s software [src-20261022-edad3e]; AkzoNobel, BASF and Henkel use Albert [src-20261013-fbc067]. *Where startups win [inference]:* molecules incumbents have offshored (generic APIs, KSMs), small-batch custom work that big plants find uneconomic, and new AI-designed molecules owned outright (Osmo).

Regulation and qualification timelines. - **TSCA, new molecule (PMN):** 90 days by law, but the median decision took **520 days in 2025** and 469 in Q1 2026; about 92% of pending cases are past the deadline [src-20261027-f4abbf]. EPA met the 90 days less than 10% of the time [src-20250101-74b3e0]. - **TSCA, low volume (LVE, up to 10,000 kg a year):** 30-day review, the usual path for specialty pilot volumes [src-20261027-f4abbf]. - **TSCA, existing molecule made a new way:** no new-chemical notice; review applies to new chemicals and new uses [src-20261027-f4abbf][src-20261001-d40423]. *Meaning [inference]:* a new AI-designed molecule should budget about 1.5 years of EPA review before full-volume US sale. - **FDA (APIs):** a standard generic application has a 10-month review goal [src-20261027-711e48], plus cGMP plant inspection (Phlow's 2027 timing [src-20260209-16223c]). - **Pesticides:** new actives typically take 10–15 years and about $300M from discovery to market (**Claim, Enko CEO**) [src-20250601-f7467b]. - **EU REACH:** registration applies before selling into Europe [src-20251110-84b67e][src-20251218-f0d6ef].

5-year outlook (to 2031). *Pharma APIs and intermediates:* **possible**. Phlow could supply in 2027 if FDA approves; policy creates buyers. *Fragrances and custom specialty:* **likely** to be selling at meaningful scale (Osmo, Scimplify already sell). *Agrochemical actives:* **unlikely** (registration timelines). *AI-designed new molecules at volume generally:* **unlikely** outside fragrance, given TSCA and qualification. **Bear case, equal weight:** see 5.11. US specialty demand is flat, generics are outside the tariff, incumbents are qualified and buying AI tools, and the first AI-designed molecules sell mainly to small customers.

6. Advanced Materials

> **Our view: Advanced Materials** *(analyst judgment, not fact [inference])* - **Stance:** Selective. Back producers with an urgent buyer; watch discovery labs. - **Conviction:** low to medium. The needs are new (PFAS-free cooling, hypersonic alloys) but no AI-designed material sells at scale yet. - **Why:** (1) only the plant owner captures the margin at the supply step (6.4); (2) startups win where no incumbent is qualified (6.14); (3) liquid cooling is expected to reach 53% share in 2026 [src-20260817-13658d]. - **Strongest counter:** qualification takes 1–5 years and can outlast runway; Chinese cathode prices cap margins; investors say there is "no AlphaFold moment" for materials yet [src-20260528-2c20ee]. - **What would change our mind:** Orbital's fluid shipping with 2027 GPUs; a discovery lab selling material, not papers. - **Best placed / watch:** Orbital, Mitra (fully integrated); Radical (hybrid); CuspAI, Periodic, Lila (enablers). - **Good SPC fit:** Series A–B; makes its own product, or uses a contract maker while keeping the margin; buyer is a data-center, defense or battery customer already in qualification.

What this industry is. Advanced Materials covers companies that make and sell engineered materials rather than commodity molecules: battery cathodes, cooling fluids and hardware for data centers, sorbents, alloys and catalysts. Two producers sit here (Mitra Chem, Orbital Industries). AI discovery labs (Radical AI, CuspAI, Periodic Labs, Lila Sciences) are kept in a clearly labeled sub-group, **Materials Discovery Labs**: they invent materials with AI and robotic labs but mostly do not yet sell product at scale.

6.1 The problem today

- **China makes almost all battery materials.** In 2025 China made 70% of the world's electric vehicles, more than 80% of battery cells, about 85% of cathode active material and more than 90% of anode material [src-20260501-79ef76]. For the cheaper LFP (lithium iron phosphate) chemistry, more than 98% of cathode material and cells are made in China [src-20250521-9be25a].

- **The US is switching to the chemistry it makes least.** LFP was more than 55% of EV batteries worldwide in 2025. In the US, more than 50 GWh of cell capacity (LG Energy Solution, Ford) has been reassigned to LFP, much of it for grid storage, which was a third of US battery deployment; LFP cathode and its precursors are almost entirely made in China [src-20260501-600ef9].
- **Data centers need new cooling materials.** Data centers used 415 TWh of electricity in 2024 (1.5% of the world total), heading to about 945 TWh in 2030, growing about 15% a year; cooling takes from 7% (hyperscale) to over 30% (enterprise) of a data center's power [src-20250410-22e27a]. Liquid cooling is expected to reach 53% of AI chips in 2026 (med) [src-20260817-13658d].
- **The old fluorine chemistry is being regulated out.** Many cooling fluids, seals and coatings use PFAS ("forever chemicals"). EU scientific committees backed a broad PFAS restriction in March 2026, with targeted exemptions; fluoropolymers and fluorinated gases are in scope [src-20260326-379e42].
- **Discovery is cheap, scale-up is not.** AI shortens the first two steps of making a new material (weeks to months), but scale-up and customer qualification take one to five years each (6.6).

What this means [inference]: there is real, policy-backed demand for a non-Chinese cathode and for new cooling materials, but the winner is whoever can make tons at a qualified spec, not whoever finds the molecule first.

6.2 Macro and tailwinds, ranked [ranking is inference]

Rank	Tailwind	Evidence	Durable?
1	**Battery supply-chain localization**	China ~85% of cathode, >98% of LFP cathode [src-20260501-79ef76][src-20250521-9be25a]; Mitra has a $100M DOE award for a Michigan plant [src-20261023-6e0642]	Structural (security), but subsidy-dependent
2	**Grid-storage pivot to LFP**	>50 GWh of US capacity moved to LFP; storage a third of US deployment [src-20260501-600ef9]	Structural while data centers add load
3	**Data-center power and cooling**	415 → ~945 TWh by 2030; cooling 7–30%+ of power [src-20250410-22e27a]; liquid cooling 53% of AI chips in 2026 [src-20260817-13658d]	Structural; tied to AI capex
4	**PFAS rules**	EU restriction backed March 2026 [src-20260326-379e42]; US keeps 4 ppt PFOA/PFOS limits, deadline 2031 [src-20260520-caa33e]	EU structural; US partly reversed
5	**AI and lab automation getting cheaper**	Discovery labs funded (Periodic, Lila, CuspAI; 6.10); open datasets like OMol25 (100M+ calculations) [src-20250513-fbb4d8]	Structural (tech), hype-sensitive

Battery chain, in more depth. The IEA expects China still to hold about three-quarters of cathode and about 90% of anode production in 2035; the EU and US together reach only about 10% of cathode [src-20260501-79ef76]. China's edge in LFP comes from cheap inputs, equipment and patents [src-20250521-9be25a]. *Meaning [inference]:* a US cathode maker sells mostly security of supply, so its

customers are those who must or will pay for it (automakers with tax-credit rules, defense, utilities buying storage).

Cooling, in more depth. Accelerated (AI) servers are the fastest-growing slice of data-center load, and the US alone accounts for almost half of its electricity demand growth to 2030 from data centers [src-20250410-22e27a]. Each point of cooling efficiency is power that can go to chips instead [inference]. Orbital targets exactly this with fluids and hardware, with qualification aimed at 2027 [src-20260528-375aec].

PFAS, in more depth. The EU's restriction would cover manufacture, sale and use with time-limited exemptions [src-20260326-379e42]. In the US, the EPA kept the 4 parts-per-trillion limits for PFOA and PFOS in drinking water but proposed moving compliance from 2029 to 2031 and rescinding limits for four other PFAS (May 2026) [src-20260520-caa33e]. *Meaning [inference]:* PFAS-free replacements have a firm market in Europe and a softer one in the US.

6.3 Headwinds, at equal weight

1. **Chinese price.** China's scale sets cathode prices; a US plant is a price-taker without offtake or subsidy [src-20250521-9be25a] [inference].
2. **EV slowdown in the US.** US LFP share in EVs almost halved; capacity is being re-pointed to storage [src-20260501-600ef9].
3. **Scale-up and qualification take years** (6.6) and are where most money is spent.
4. **Rules can soften.** The US PFAS rollback shows regulation-driven demand can shrink [src-20260520-caa33e].
5. **Discovery-lab hype.** Large rounds before revenue (6.10); Plural, an Orbital investor, says there is "no AlphaFold moment" in materials [src-20260528-2c20ee].

6.4 Why an AI-enabled, vertically integrated company is the answer here

Owning the plant is the only way to capture the margin in step 5 (supply) of the materials chain; a lab that licenses an invention captures little [inference]. AI helps most where a producer can run many experiments on its own line: Mitra uses AI from lab scale (1 kg/day) to production (1,000 kg/day) [src-20261023-6e0642]. **Where it breaks:** discovery speed does not shorten customer qualification, and Chinese cost still sets price.

6.5 What would have to be true, and signals to watch

- A US cathode plant reaches steady output at a price buyers accept. **Watch:** Mitra's Muskegon plant start-up and offtakes [src-20261023-6e0642].
- Liquid cooling stays standard as AI chips grow. **Watch:** liquid-cooling share (53% expected in 2026) [src-20260817-13658d].
- The EU PFAS restriction is adopted. **Watch:** the final SEAC opinion and Commission proposal [src-20260326-379e42].
- A discovery lab sells product, not just papers. **Watch:** Periodic, Lila, CuspAI revenue from materials.

6.6 Macro in more depth

The steps, with typical time in brackets [inference, built from sources cited]: 1. **Compute screen** (weeks): the model narrows millions of candidates to hundreds. Now cheap. 2. **Synthesis and first tests** (months): grams to 500 g in an SDL. This is the step labs speed up 6–17 times. An RSC benchmark finds a median speedup of about 6 times over random sampling (search; low), and Lila claims 17 times [src-20260925-679418]. 3. **Scale-up** (1–3 years): kilograms to tons. It needs process

chemistry or metallurgy, and a pilot line. Radical's RAI-939 is at this step now [src-20260807-52b690]. 4. **Qualification** (1–5 years): the customer tests the material in its own product against specs. Aerospace alloys need allowables databases. Chip fluids need qualification by the fab or GPU maker (Orbital, 2027 target [src-20260528-375aec]). Water sorbents need regulatory approval (Kemira). 5. **Supply** (ongoing): someone has to make it reliably at cost. This is where the margin is, and where owner-producers sit.

Where AI helps: step 1 a lot, step 2 a lot, steps 3–5 little. Steps 3–5 are most of the time and money. That is why Plural says there is "no AlphaFold moment" [src-20260528-2c20ee], and why the Lila CTO says only about 5% of science is cheap to verify [src-20260917-29aa29].

Materials discovery, step by step.

A new material (an alloy, a battery cathode, a catalyst, a coolant, a magnet) normally takes 10–20 years and tens of millions of dollars to go from idea to qualified product. Discovery labs try to compress that with three tools: 1. **Generative and predictive models.** Software proposes candidate structures and predicts their properties, so you test hundreds instead of millions. Examples: - Microsoft's MatterGen is a diffusion model trained on about 608k stable materials. It generates crystal structures to a target property [src-20250116-63e766][src-20261022-3e2a9d]. - Google DeepMind's GNoME predicted 2.2M crystals, about 380k of them stable. Outside labs have since made 736 of them [src-20231129-1dc706]. - Meta's open OMat24 dataset holds more than 110M quantum-mechanics (DFT) calculations [src-20130101-5e5543]. 2. **Self-driving labs (SDLs).** Robots mix, synthesize and test the candidates, and the results feed back into the model. Examples are Berkeley's A-Lab, Argonne's Polybot [src-20250217-059784] and Radical AI's alloy lab. 3. **The closed loop.** Model proposes, robot makes and measures, model updates. The claim is that each turn makes the next guess better, so the lab gets faster over time. This is the materials version of the "data loop" in section 6.

The key point [inference]: models are becoming free. GNoME, MatterGen, OMat24 and the Genesis Mission platform are all public or close to it. The scarce assets are (a) physical throughput, meaning how many real samples you can make and test, (b) data that is proprietary and tied to real-world performance, and (c) the ability to scale up and qualify a material with a customer. That is where the right to win sits.

6.7 How the industry and its supply chain work

Model	What they sell	Examples	Producer test?
1. Software / SaaS	Data platform and AI tools to the R&D teams of chemical and materials companies	Citrine, Uncountable, Kebotix (status unclear), Albert Invent, Microsoft Discovery	No
2. Discovery-as-a-service / IP	Custom discovery projects, licensing and royalties, with partners making the material	CuspAI, Periodic, Lila (materials arm), Aionics, Dunia	No
3. Lab plus government / partner contracts	Its own SDL; paid by defense and labs, then licensing or making specialty material	Radical AI	Partly (makes alloys at small scale)
4. Owner-producer	Designs the material **and** makes and sells the product	Orbital Industries, Mitra Chem	Yes, closest to the thesis

Market size (this industry). No free market size for materials R&D in the repo [gap]. Physical proxies: China holds about 85% of cathode active material and more than 90% of anode material [src-20260501-79ef76]; data centers used 415 TWh in 2024, heading to about 945 TWh by 2030, with cooling 7% to over 30% of that power [src-20250410-22e27a].

6.8 Where AI and automation help

AI screens candidates and runs robotic labs (steps 1–2 in 6.6); producers use it to tune their own lines, as Mitra does from 1 kg/day to 1,000 kg/day [src-20261023-6e0642].

6.9 Right to win, unit economics and data loop

Right to win [inference]. Make the material at a qualified spec in the US, with an offtake; discovery alone captures little margin.

Data loop (score 0–3; scale in 2.13):

Company	Score	How the loop works	Best evidence / what is missing
Radical AI	2	Robot lab tests and learns in a closed loop	8-20 alloys a day [src-20260617-f9d293]. Missing: hit-rate trend
Periodic, CuspAI, Mind	1	Lab or factory as a training ground for AI	[src-20260917-1bcb81][src-20261022-37a0ea][src-20260513-83aa83]. Missing: production evidence

6.10 Companies, grouped by label

Producers — fully integrated

> **Our view: Advanced Materials producers** *(analyst judgment)* — **Selective; Orbital first. Conviction: medium. Why:** Orbital has an urgent buyer (AWS) and owns the product margin [src-20260528-375aec]; Mitra uses AI from lab to production [src-20261023-6e0642]. **Counter:** both are pilot-stage, and Mitra competes with Chinese LFP on cost per kWh, the hardest market in industry. **Change our mind:** Orbital's fluid qualifies for 2027 GPUs; Mitra's plant is funded with offtakes. **Best placed:** Orbital, Mitra (fully integrated). **Good fit:** a qualified buyer, a contract maker or own line, and a plan for gross margin per unit.

Orbital Industries (formerly Orbital Materials; London and Princeton, NJ; model 4). - **Funding:** $50M Series B led by Plural, May 28, 2026; about 50 staff [src-20260528-93befb][src-20260528-375aec]. - **Model:** it is vertically integrated and sells products, not software. The two products are a PFAS-free dielectric cooling fluid for data centers and "Nova Array," a modular data center it says can be deployed in 180 days [src-20260528-93befb]. - **What it means [inference]: the most thesis-aligned name in the section.** It found a buyer with an urgent need (data-center cooling), chose a product it can make through a contract manufacturer, and owns the margin. If the fluid ships in 2027 it could be among the first AI-designed materials sold at scale. Diligence: who the CMO is, qualification status, gross margin per liter. - **v9.9 (company claim, CEO Jonathan Godwin):** its cooling fluid took about 3 years from start to market, vs a new material that "would normally cost a hundred million dollars and ten years" [src-20260629-ae1f90, 3:09].

Mitra Chem (Mountain View, CA; LMFP and LFP battery cathodes; model 4). - **History:** began shipping LMFP cathode samples to global customers in 2023 [src-20240611-24c650]. Series B led by GM. - **Production: Per BusinessPost (Korean):** a $100M DOE production grant, $50M from Michigan's MSF fund, and a planned 30k-ton-a-year, $370M LFP plant in Michigan. Korea's L&F invested $10M for about 3.3%, and the two are considering a joint-venture plant [src-20260831-db47e6] (med; single foreign-language article). - **What it means [inference]:** an AI-designed chemistry turning into a chemical plant, so it does pass the producer test. But it competes on cost per kWh against Chinese

LFP, which is the hardest market in industry. Its value depends on 45X and the foreign-entity rules (section 2.3).

Materials Discovery Labs (sub-group, kept separate from producers)

> **Our view: Materials Discovery Labs** *(analyst judgment)* — **Watch, not invest yet. Conviction: medium (that they are early). Why:** labs are priced on talent (Periodic about $7.5B reported) and earn early revenue mainly from on-site services to one unnamed chipmaker [src-20261010-7b9ecf]; none sells material yet (6.14); discovery speed does not shorten qualification (6.4). **Counter:** CuspAI has about 30 industry partners [src-20261022-37a0ea] and Radical has a defense buyer, so some may become producers. **Change our mind:** a lab sells material or signs a supply contract. **Watch:** Radical (hybrid), CuspAI, Periodic, Lila (enablers). **Good fit:** a lab with a defense or industrial buyer and a partner to scale up production, like Radical.

Radical AI, CuspAI, Periodic Labs, Lila Sciences and smaller names. They invent materials; they do not yet sell product at scale.

Pure discovery labs and design partners. Most do not run a production floor, so they are tracked, not targeted (section 6). Labels: Radical AI hybrid; CuspAI, Periodic and Lila enablers.

Radical AI (New York; alloys and metals; model 3; Hybrid; partial fit). - **Funding:** $55M led by RTX Ventures [src-20250718-d46f24]. An Air Force AFWERX Direct-to-Phase II contract worth $1,197,902 [src-20250825-300a14]. - **Technology:** its AI engine, “Antimatter,” drives a self-driving alloy lab. TIME named it one of the Best Inventions of 2026 [src-20261020-055082]. - **What it means [inference]:** the clearest defense-led path. Hypersonic and propulsion alloys have a buyer (RTX and the Air Force) and a clear test (the torch test). The risk is the jump from a 500 g button to qualified mill product, which needs a melter and forger partner. - **v9.9 (company claims, CEO Joseph Krause):** about 1,200 alloys made in five or six months; it has made a hafnium-free version of the C103 aerospace alloy after hafnium prices rose 10–15× on Chinese supply; “we haven’t touched manufacturing”; “models aren’t the moat, experiments are” [src-20260617-f9d293, 9:11, 9:44, 16:33–17:08, 1:14:26].

CuspAI (Cambridge, UK; sorbents, membranes, chemicals; model 2; Enabler; track). - **Funding:** $450M Series B (Kleiner, NEA) [src-20260720-840ae3]. - **Partners:** about 30 industry and 15 lab partners, including Samsung, AMD, Lam and Applied [src-20261022-37a0ea]. Its “AI Materials Foundry” consortium has 45+ members. - **What it means [inference]:** the best commercial traction in the group, with a named buyer (Kemira) and a regulatory driver (PFAS rules). But CuspAI is a design shop: Kemira makes and sells the product, and CuspAI’s take is a royalty or fee.

Periodic Labs (San Francisco; semiconductors, superconductors, magnets; model 2; Enabler; track). - **Funding:** $300M seed, September 2025 [src-20251022-8eb7c0]; reportedly raising more (med) [src-20260514-5796ab]. - **Go-to-market: Claim (Periodic, via Heard In):** acts as its own “customer zero.” It sends forward-deployed engineers, runs models on the client’s site and fine-tunes on the client’s data. Its first focus is semiconductor heat dissipation [src-20261008-38b3d6]. Its website lists magnets, semiconductors and superconductors, plus a Montreal ML office [src-20261001-e0573e][src-20261007-13c943]. - **What it means [inference]:** the most “frontier lab” of the group (OpenAI and DeepMind alumni). It is priced on talent; commercial proof is the least public.

Lila Sciences (Cambridge, MA; “scientific superintelligence,” life sciences and materials; model 2; Enabler; track). - **Funding and site:** $550M raised and a new 250k sq ft site in Cambridge [src-20260610-21e4df]. Reportedly in talks at about $8.5B (Bloomberg, via repo notes). - **Catalysts (company claim):** 2,942 catalysts screened in 3 months, across six palladium-based families. The

best matched ruthenium after more than 1,000 hours of testing. 17 times faster, with over 90% less hands-on time [src-20260925-679418]. Still untested in a real electrolyzer [src-20261019-392916]. - **What it means [inference]:** most of the value is in biology. Materials is a second line. Not an industrial producer.

6.11 Bear case, at equal weight

1. **No AI-designed material is clearly in volume commercial sale yet** [inference, from sources]. Orbital targets 2027 [src-20260528-375aec]. Kemira's PFAS sorbent and RAI-939 are at pilot or test-article stage [src-20260521-f4b51b][src-20260807-52b690].
2. **The science record is mixed.**
    - A-Lab's claim of 41 new materials was challenged ("no new materials were discovered") [src-20240124-513fd9].
    - For GNoME, the 736 independent syntheses are under 0.2% of the 380k predicted-stable crystals [src-20231129-1dc706].
    - MatterGen is limited to about 20 atoms per unit cell [src-20261022-3e2a9d].
3. **Synthesis and qualification, not discovery, are the bottleneck.** AI speeds up the cheap part.
4. **Buyers have small budgets and refuse to share IP.** R&D software sells in the hundreds of thousands; majors refuse royalties [src-20260913-3ad55e][src-20261016-29997e].
5. **The models are commodities:** open models and Genesis.
6. **Valuations far ahead of revenue:** CuspAI ($450M raised), Periodic ($300M seed), Lila (talks at about $8.5B, reported), none with disclosed material revenue [inference].
7. **Legal and safety issues:** the Royal Society review notes that patent law recognizes only human inventors, and flags safety and cyber risks of autonomous labs [src-20250716-e94853].
8. **Market size:** the lab-automation market is about $8–9B today, perhaps $20–24B later [src-20260709-1159ab] (low). That is small next to manufacturing.

6.12 Key sources

- IEA Global EV Outlook 2026 [src-20260501-79ef76][src-20260501-600ef9]; IEA LFP supply chain [src-20250521-9be25a]
- IEA Energy and AI [src-20250410-22e27a]; TrendForce liquid cooling [src-20260817-13658d]
- ECHA PFAS [src-20260326-379e42]; EPA PFAS [src-20260520-caa33e]
- Mitra [src-20261023-6e0642]; Orbital [src-20260528-375aec][src-20260528-93befb][src-20260528-2c20ee]
- Discovery labs [src-20260807-52b690][src-20260925-679418][src-20260917-29aa29]; OMol25 [src-20250513-fbb4d8]

6.13 Listen & watch

Full list: Appendix D.

Producers

1. ★ **"Jonathan Godwin, CEO at Orbital: Orbital raised $50m Series B"**, *Scaling Europe*, Jun 29, 2026, 21 min. ▶ Open link — *Why:* Orbital's founder on why it builds and sells hardware (a GPU-cooling material) instead of only models (company claims). Not in the repo (new).
2. **"Developing a supply chain for American-made batteries"**, *Volts*, Sep 6, 2024, length not listed (free transcript on page). ▶ Open link — *Why:* Mitra Chem's CEO on China's grip on LFP cathodes and its iron-rich roadmap. Not in the repo (new).

Materials Discovery Labs

1. ★ **“The Self-Driving Lab — Joseph Krause, Radical AI”**, *Latent Space*, Jun 17, 2026, length not listed. ▶ Open link — *Why:* the moat is experiments, not models; customers asking for 300 lb pushed Radical toward making alloys. Repo: src-20260617-f9d293.
2. **“AI for Atoms: Periodic Labs with Liam Fedus”**, *No Priors*, Apr 3, 2026, 29 min. ▶ Open link — *Why:* economics of an AI lab running its own experiments. Repo: src-20260403-40fa0e.
3. **“Meet the Ex-OpenAI & DeepMind Leads Creating an AI Scientist”**, *The Generalist*, Sep 29, 2026, 60 min. ▶ Open link — *Why:* Periodic’s superconductor focus and the impurity problem. Repo: src-20260929-ca6d19.
4. **“Max Welling, CuspAI — Nebius Conversations”**, May 4, 2026, 19 min. ▶ Open link — *Why:* the software-only model for contrast; it designs but does not make. Repo: src-20260504-36655f.

6.14 Commercial traction and incumbents

Labels as in 3.14.

Who’s actually commercializing: detailed traction (researched Oct 27, 2026; free sources)

*Ranked from most to least commercial [ranking is inference]. Model: M = sells material, L = licenses IP, S = R&D services, SW = software. Company statements are **Claims**.*

#	Company (label)	Customers / partners and what they buy	Contracts and money	Product, capacity, hiring	Model	Stage	Strongest evidence / gap
1	Citrine Informatics (Enabler)	Huntsman, Perstorp (alkyd emulsion R&D) [src-20261022-267188][src-20260101-bf65ea]; multi-year LyondellBasell deal [src-20260902-224ead]	**Claim:** 10 customers on multi-year licenses [src-20260215-cf8e3f]	Software platform; no material	SW	**Early revenue**	Named chemical majors paying for software / no audited revenue; sells tools, not materials
2	Orbital Industries (Full)	AWS: strategic partner, pilots cooling and carbon-removal tech [src-20260528-375aec]; BASF licenses CurieOS [src-20261022-edad3e]	$50M Series B [src-20260528-93befb]	Cooling fluid in qualification for 2027 GPUs; made through a contract maker	M + SW	**Pilot**, some licensing revenue	AWS and BASF named / no fluid sold yet; contract maker not disclosed
3	Aionics (Enabler; not previously in primer)	Cellforce (Porsche), Form Energy, Resonac, Cuberg: custom electrolyte designs [src-20240101-41fffd]	Not disclosed	Electrolyte formulations; no own plant	S / L	**Early revenue** (services)	Named battery customers / older source (2024); no volumes or contract values

#	Company (label)	Customers / partners and what they buy	Contracts and money	Product, capacity, hiring	Model	Stage	Strongest evidence / gap
4	Mitra Chem (Full)	**Claim:** LFP under qualification with “top EV and ESS customers”, none named [src-20261023-6e0642]; Sun Chemical (manufacturing partner) and L&F (investor, planned plant partner) [src-20260115-864254]	$100M DOE grant, retained May 2026; $50M Michigan [src-20260805-60cb6f]	Samples since 2023; **Claim:** 1 kg/day to 1,000 kg/day [src-20261023-6e0642]. Muskegon plant: groundbreaking early 2027, commissioning 2028 [src-20260805-60cb6f]	M	**Pilot** (samples)	Federal money retained / no named offtake; plant date slipped from “by 2027” to 2028 commissioning [src-20260805-60cb6f]
5	Chemix (Hybrid; not previously in primer)	UBCO (electric motorbikes): cobalt-free battery packs [src-20230501-de07c8]	$20M Series A (Porsche Ventures among investors); **Claim:** 2 GWh supply agreement with an unnamed EV maker [src-20240424-486875]	Cells and packs designed with AI, made by partners	M (cells via partners)	**Early revenue / pilot**	One named customer / the 2 GWh deal buyer is unnamed and sources are 2023–24
6	CuspAI (Enabler)	Kemira: AI-designed MOFs for PFAS removal; about 20 candidates moved to next phase; framework agreement for more projects [src-20260521-f4b51b]	$450M Series B [src-20260720-840ae3]; ~30 industry partners [src-20261022-37a0ea]	Designs only; Kemira would make and sell	L / S	**Pilot** (paid partnerships)	Named buyer with a regulatory driver / no material in production; royalty terms unknown
7	Periodic Labs (Enabler)	One unnamed semiconductor maker (chip heat dissipation) [src-20261001-e0573e]; on-site engineers train models on partner data [src-20261010-7b9ecf]	**Reported (low):** early semiconductor revenue by March 2026 [src-20260315-28f497]; $300M seed	Hiring forward-deployed engineers for fabs, Taiwan on-site [src-20261001-6bb427]; first academic paper not yet out [src-20261010-7b9ecf]	S / SW	**Early revenue** (services, unverified)	Hiring shows real deployments / no partner named, no material found; currently a services business

#	Company (label)	Customers / partners and what they buy	Contracts and money	Product, capacity, hiring	Model	Stage	Strongest evidence / gap
8	Radical AI (Hybrid)	US Air Force (hypersonic alloys); Sandia (hydrogen-purification materials research); Purdue (testing) [src-20250521-4c5901][src-20260929-dfb65a]	USAF Direct-to-Phase II $1,197,902, to Feb 2027 [src-20250521-4c5901]	**Claim:** 300+ alloys in 16 weeks; RAI-939 beat C103 in a torch test (Purdue) [src-20260807-52b690]; scaling to bigger test pieces	S / M (aim)	**Pilot** (government R&D)	Independent test result / only ~$1.2M of contracts; no production partner yet
9	Mattiq (Enabler; not previously in primer)	Heraeus: low-iridium electrolysis catalysts toward production [src-20260101-dea6ea]	ARPA-E CATALCHEM-E participant (LinkedIn, unverified)	Chip-based “megalibrary” screening	S / L	**Pilot**	Heraeus is a credible partner / no revenue disclosed
10	Lila Sciences (Enabler)	Caltech, LBNL, Northwestern, Argonne, NLR: research partners [src-20260722-5412a7]	DOE Genesis Mission: three Phase I awards (value not disclosed) [src-20260722-5412a7]; $550M raised	Hiring enterprise sales for materials, chemicals, energy and semiconductor accounts [src-20261006-732675]; catalyst screen still untested in an electrolyzer	S / SW	**Lab** (selling begins)	Sales hiring signals intent / no named commercial customer
11	Dunia Innovations (Enabler)	Unnamed customers in catalysts, batteries, semiconductors [src-20260520-57be93]	GigaLab €280M, due 2028	Robotic lab	S	**Lab**	Capacity plan / no named customer
12	Entalpic (Enabler; not previously in primer)	ATLANT 3D (digital twins for ALD reactors); IFPEN research [src-20260301-4df637]	Nvidia investment, Oct 2026 [src-20261015-cfa201]	Software and simulation	SW	**Lab**	Nvidia backing / **pivot:** WSJ says it put its climate work on ice [src-20261015-cfa201]
13	Discovered Materials (Enabler)	None named [src-20260810-357833]	$9M seed	Candidate materials released	SW / L	**Lab**	/ no customers
14	Kebotix (Enabler; not previously in primer)	None current	$5M round Aug 2024 [src-20240805-08be84]	Self-driving lab	S	**Lab, possibly stale**	Website live / no 2025–26 news found; treat as stalled

Checked and excluded: Polaris (an NSF-funded university lab at UChicago/Argonne, not a startup) and "Matter Labs" (the University of Toronto Matter Lab is academic; GrAI Matter Labs makes chips). Neither is a company to track.

Narrative [inference]. No AI materials company sells a new material at volume yet. The money is earned in three ways: software licenses (Citrine, Orbital's CurieOS), paid design work for a named industrial partner (CuspAI–Kemira, Aionics, Mattiq–Heraeus), and government R&D (Radical, Lila). Periodic's reported revenue is on-site services in fabs, which looks more like consulting than discovery. The two producers, Orbital and Mitra, are closest to selling material, but both are gated by qualification and Mitra's plant has slipped to 2028. **Bear case, equal weight:** design work for partners is small and the partner keeps the margin; government awards are around $1M; the best-funded labs ($300M–$550M) have the thinnest named commercial evidence. **Weird flags:** Entalpic's pivot away from climate; Kebotix quiet since 2024; Mitra's plant timing slipped; Chemix's 2 GWh buyer is unnamed.

Revenue vs lab. Producers are at pilot or sample stage; the only clear revenue is software and licensing (Citrine, Orbital's CurieOS). Discovery labs are lab-stage: investors say there has been "no AlphaFold moment" for materials yet [src-20260528-2c20ee].

Incumbents. Albemarle ($5.14B net sales in 2025, $2.71B from energy storage) [src-20260201-955851]; Umicore (€3.6B revenue; its battery-materials unit had €436M revenue and still lost money) [src-20260220-f497c9]; Corning (a qualified supplier on defense and electronics lists) [src-20261022-1389b7]. *What they do better:* qualified products, long customer trust and plants already built. Umicore spent decades mastering cathodes [src-20250529-3a588b]. *Where startups win [inference]:* new needs where no incumbent is qualified (PFAS-free cooling fluids, hypersonic alloys). Even Umicore's losses show that battery materials are hard for everyone; China sets cathode prices [src-20250521-9be25a].

Regulation and qualification. Customer qualification takes 1–5 years (6.6). Battery products also need safety certification; full UL compliance typically takes 12–20+ weeks (low; vendor source) [src-20250101-040756]. New chemistries also face TSCA review (5.14).

5-year outlook (to 2031). *Producers:* **possible** for one or two names (Orbital if its fluid qualifies in 2027; Mitra if its plant is funded). *Discovery labs:* **unlikely** to be selling material at scale. **Bear case, equal weight:** see 6.11. Qualification can outlast a startup's runway, Chinese cathode pricing caps margins, and discovery labs may end up as royalty businesses, not producers.

7. Enabling Software & Robotics (tracked, outside the core)

> **Our view: Enabling Software & Robotics** *(analyst judgment, not fact [inference])* - **Stance:** Watch, not invest yet; track as benchmarks and as tools inside producers. - **Conviction:** medium. The tools matter, but selling them alone is a smaller, more competitive business (7.4). - **Why:** (1) traction is first-customer stage (Arda: Anduril only [src-20261020-a1e174]; Mind: Rivian [src-20261013-cc78c0]); (2) GrayMatter pivoted to robots as a service after software-only sales did not work [src-20261022-3335bd]; (3) incumbents own the installed base. - **Strongest counter:** an enabler that becomes the standard for thousands of small factories could beat owning plants (Arda's bet). - **What would change our mind:** retention data from Arda; robot success above ~90% in real work; producers licensing their software (MasonOS, MarianaOS). - **Best placed / watch:** Arda, Mind, GrayMatter, Lumaril (enablers). - **Good SPC fit:** only where it is tied to owned capacity, e.g. robot cells as a service with usage pricing.

What this industry is. Enabling Software & Robotics covers companies that sell tools to factories rather than making the product themselves: factory operating systems, inventory and ordering software,

inspection AI, CAM automation and robots. We track it because it sets the pace for the producers in chapters 3–6, but it sits **outside the core thesis** (it is an enabler, not an owner of output).

7.1 The problem today

- **Factory workflows are slow and manual. Claim (Accel, on Arda):** assembly instructions drop from hours to minutes [src-20261020-8d7abc].
- **The US is under-automated.** A record 603,307 robots were installed worldwide in 2025; China installed 354,226 and Chinese brands supply over half of China's installs; the US installed 38,428 [src-20260924-28b59b], only about 3,000 in metal and machinery, at a density of 307 per 10,000 workers versus 1,220 in South Korea [src-20260618-c903cc].
- **Robots still fail often:** robot-learning success is 59–66% on short tasks versus 99.9%+ needed [src-20260904-267293].

7.2 Macro and tailwinds, ranked [ranking is inference]

Rank	Tailwind	Date / size	Structural or reversible?
1	**Labor shortage**	2.7	Structural
2	**Cheaper robots and models**	Robot cost vs $46.30/hr loaded labor (2.11) [src-20260826-14e4bb]	Structural
3	**Manufacturing expanding**	ISM 54.5 (Sept 2026) [src-20261008-89df84]	Cyclical
4	**100% bonus depreciation**	[src-20260114-2075f7]	Reversible
5	**Value shifting to software**	Roland Berger: automation growth of 6–7% a year to 2030 (up to 9%), value moving from hardware to software, pharma strongest [src-20260116-0c4980] (med)	Structural

- **Incumbents capture most near-term spend.** Rockwell's sales rose 8% with a 49.5% gross margin [src-20260804-0e8130]; Siemens bought Altair [src-20250326-cc1810].
- **Budgets are tight.** NAM: 78.9% positive, but capex growth expected at only 2.6% and 63.0% plan to import machinery [src-20260914-b4d55c]. That favors tools that pay for themselves from output [inference].
- **Process plants pay for throughput.** Chemical assets need about 80–85% utilization to earn their cost of capital; Europe averages about 74% [src-20260401-e2adb0][src-20260915-e74614]. In a sold-out plant each extra percent of yield is near-pure margin [inference], the opening for closed-loop control in chemicals, polymers and biologics.
- **How far costs can move.** BCG: up to 60% productivity gains and 30–60% less labor in some plants, but greenfield aerospace and defense still favors China [src-20260528-a1c5d1].

7.3 Headwinds, at equal weight

1. **Small factories buy slowly** and churn [inference].
2. **Incumbents bundle software** with hardware (Siemens, Rockwell) [src-20250326-cc1810][src-20260804-0e8130].
3. **Reliability gap** in robot learning [src-20260904-267293]; **Claim (Mehr):** physical AI is at "2007-2008 of LLMs" [src-20260220-0d5345].
4. **Rates and capex caution** (2.5).
5. **Value capture:** a vendor never owns the output margin, and owner-operators may build their own software [inference].

7.4 Why vertical integration matters here, and where the logic breaks

- Software and robots are the **means** by which the producers in chapters 3–6 win; selling them separately is a smaller, more competitive business [inference].
- An enabler can still win as the standard for thousands of small factories (Arda's bet).
- **Where it breaks:** if owner-operators build in-house, the enabler market shrinks to slow-buying job shops.

7.5 What would have to be true, and signals to watch

- Usage- or outcome-based pricing sticks. **Watch:** Arda and Lumaril retention (claims only so far).
- Robot success rates rise above ~90% in real work.
- Producers license out their software (MasonOS, MarianaOS).

7.6 Macro in more depth

AI makes the office work around the machine (quoting, programming, instructions) faster today; it does not yet run machines on its own [inference]. LLM agents read CAD and write work instructions (Arda: 10 hours to 15 minutes at Anduril) [src-20261020-914a64]; Freeform runs GPU clusters on site to control each printed layer [src-20260219-2125d7]. **Claim (Nox CEO):** "AI just doesn't know so much of the nuance" of a factory [src-20260325-5b68db]. Data loops in general: 2.13.

7.7 How the industry and its supply chain work

- **Who:** Arda (CAD to work instructions) [src-20261020-914a64]; Mind Robotics (robot AI; "a partner of Rivian, but not a spinoff") [src-20261013-cc78c0]; candidates Rapta, Allus, Faraday. Incumbents: Siemens (bought Altair and Dotmatics) [src-20250326-cc1810][src-20250701-a7fef9], Rockwell.
- **Risks:** incumbents own the customers and the host plant owns the data; value needs connected plant data and new workflows [src-20261006-6265ad]. Mind has raised over $1B and shown no robot publicly [src-20261013-cc78c0]. Bright Machines pivoted to assembling AI servers [src-20240926-fd90fb].
- **Enablers arm the incumbents.** GrayMatter found selling autonomy software "doesn't really work" and now rents turnkey cells "as a service"; **Claim:** Riddell went from about 300 to 1,700 helmets a day on the same robots [src-20261022-3335bd]. CloudNC's CAM Assist writes machining plans inside GibbsCAM [src-20261021-9769d1][src-20261009-7cafd0]; AI Robotics inspects welds on 24 lines [src-20261011-a8b14d]. This layer may help old shops catch up as much as new ones [inference].
- **Market size.** Robot installs $18.6B (2025) [src-20260924-28b59b]; closed-loop process-control software has no free size [gap].

7.8 Where AI and automation help

Work instructions from CAD, inventory automation, vision inspection, robot programming, CAM automation and closed-loop process control.

7.9 Right to win, unit economics and data loop

Company	Loop score (0–3)	How it works	Missing
Arda	1	Agent improves with host's design data [src-20261020-914a64]	Who owns the data

Company	Loop score (0–3)	How it works	Missing
Lumaril	n/a	Software moves setpoints itself, so it learns from every change (claim) [src-20261027-d5ba3f]	Any customer result

7.10 Companies, grouped by label

Enabler

Lumaril (enabler; AI closed-loop control for process plants; fuller profile)

- **What it does (company claims) [src-20261027-d5ba3f].** Connects to a plant's existing controls (plus cheap audio and video sensors, "no rip-and-replace"), builds a real-time model, and changes setpoints and control logic itself through a safety layer to raise throughput and yield, in chemicals, biologics, polymers, oil and gas and manufacturing. "We're paid on the production we add." Lux describes it as codifying operator expertise [src-20261027-311fa6].
- **Team.** Co-founders Mark Elliot (creator of Palantir Foundry, Palantir's commercial CTO) and Ben Duffield (Foundry co-creator) [src-20261027-d5ba3f].
- **Funding.** Founded 2025; $12M seed led by Lux (low) [src-20261003-496a99]; a WSJ Pro headline reports $66M at a $400M valuation (article paywalled) [src-20261027-79ddd6]. Customers and results not public [gap].
- **Why it is the closest software play to the thesis [inference].** It is paid like a producer (on added output, not per seat) and runs the plant in closed loop rather than advising a human, so it gets the producer's margin and data loop without owning the plant. That lets SPC back the thesis in process industries where a new plant costs hundreds of millions.
- **Bear case.** The plant owner owns the asset and can end the contract; Honeywell, AspenTech and Emerson already sell advanced process control; plants rarely let outside software move setpoints, and one incident can end a deployment; outcome pricing invites baseline disputes; no public results [inference][src-20261003-496a99].
- **Next step [inference].** One plant's before/after yield, how the pricing baseline is set, and how the safety team approved closed loop.

Arda (factory software for manufacturing engineers; enabler / software; does not fit, competitor and comparable)

- **What it does:** an engineering workspace with an AI agent that reads CAD, works out assembly sequence and stations, and writes work instructions; it updates the line when the design changes [src-20261020-a1e174]. **Claim (CEO Bob McGrew):** retraining a human takes a day, "but it takes three to six months in order to change the automation" [src-20261020-9be48d].
- **Customers:** Anduril is the only named user, inside ArsenalOS. **Claim:** instructions from about 10 hours to 15 minutes; Anduril's COO: "We would be behind on our production goals without it" [src-20261020-a1e174]. **Claim (Altimeter):** also works with data-center transformer makers [src-20261020-a1e174].
- **What it means [inference]:** not a target (it makes no product), but a tool that could give incumbents the same speed edge, and a valuation comparable.

Mind Robotics (factory robotics AI; enabler; does not fit)

- Trains robot AI for factory tasks (picking, fastening, plugging in connectors) and sells robots, not parts [src-20261022-eede0b]. Rivian is partner, first customer and shareholder: "a partner of

Rivian, but not a spinoff" [src-20261013-cc78c0]. It depends heavily on Rivian and has raised over $1B, with no public product [src-20260515-176e26].
- **More specifics:** Mind uses Rivian as "a live, high-volume manufacturing environment for model training"; Rivian owns 32.1% [src-20261013-58de15].
- **What it means [inference]:** sells robots and models to plants (first Rivian); a comparable for the 'physical AI' enablers, not a producer.

Mentions: CloudNC (CAM automation) and GrayMatter (robot programming) — not profiled this run.

7.11 Bear case, at equal weight

- Siemens and Rockwell bundle and own the installed base [src-20250326-cc1810][src-20260804-0e8130].
- Producers may build in-house (Hadrian Opus, Isembard MasonOS, Mariana MarianaOS).
- Robot-learning reliability 59–66% [src-20260904-267293].
- Lumaril-style closed-loop control must beat Honeywell, AspenTech and Emerson APC and win plant trust (7.10).

7.12 Key sources

- IFR World Robotics [src-20260924-28b59b][src-20260618-c903cc]; robot-learning benchmark [src-20260904-267293]
- Rockwell [src-20260804-0e8130]; Siemens [src-20250326-cc1810]; Deloitte outlook [src-20251112-669746]; BCG [src-20260528-a1c5d1]
- BEA/JOLTS openings [src-20261013-a7c995]; ISM [src-20261008-89df84]; NAM [src-20260914-b4d55c]
- Roland Berger automation 2026 [src-20260116-0c4980]
- Lumaril [src-20261027-d5ba3f][src-20261027-311fa6][src-20261003-496a99][src-20261027-79ddd6]
- Arda [src-20261020-a1e174][src-20261020-914a64][src-20261020-8d7abc]

7.13 Listen & watch

Full list: Appendix D.

1. ★ **"Building Factory SuperIntelligence with Ariyan Kabir"**, *Machine Minds*, length not listed. ▶ Open link — *Why:* GrayMatter moved from licensing software to selling robot cells as a service, which tests the thesis. Repo: src-20261022-3335bd.
2. **"Former top OpenAI employees raise capital for defense startup valued at $1.5B"**, CNBC, Oct 16, 2026, short clip. ▶ Open link — *Why:* Arda on the 3–6 months it takes to retool factory automation. Repo: src-20261020-878aa9.
3. **"Mind Robotics: Building the AI-Native Robotics Platform"**, Kleiner Perkins, May 13, 2026, 4 min. ▶ Open link — *Why:* Rivian's line as Mind's live data loop. Repo: src-20260513-abe9cf.
4. **"Big Ideas 2026: Physical AI and the Industrial Stack"**, a16z Podcast, Dec 25, 2025, length not listed. ▶ Open link — *Why:* how investors frame the software layer going into 2026. Repo: src-20251225-725014.

- *Gap:* no free Lumaril podcast or founder interview found (searched Oct 27, 2026). Use the primer profile (section 7.10).

7.14 Commercial traction and incumbents

Labels as in 3.14.

Company	Status	Evidence
Arda	**Revenue (one customer)**	Anduril is the only named customer [src-20261020-a1e174].
Mind	**Revenue (first customer)**	Rivian is its first customer [src-20261013-cc78c0].
GrayMatter Robotics	**Revenue**	Rents turnkey robot cells “as a service” after software-only sales did not work [src-20261022-3335bd].
Lumaril	**Lab / pre-revenue**	No public customers [src-20261003-496a99].

Revenue vs lab. Robot cells sold as a service have revenue; AI control software is mostly first-customer stage.

Incumbents. Rockwell and other controls vendors own the installed base [src-20250326-cc1810][src-20260804-0e8130]. *What they do better:* certified hardware and plant trust. *Where startups win [inference]:* new factories (often the producers in chapters 3–6) with no legacy system.

Regulation and qualification. Little formal regulation; the hurdle is customer qualification and, for defense customers, ITAR and cybersecurity rules (4.14).

5-year outlook (to 2031). Possible, as tools inside producers rather than standalone winners. **Bear case, equal weight:** see 7.11. Software-only models have struggled (GrayMatter’s pivot) and incumbents can bundle similar features.

8. Competition and incumbents

What incumbents earn (gross margin unless noted; the bar a startup must beat): - Proto Labs: $533.1M revenue in 2025, 44.5% [src-20260206-8bb619]. Xometry: 39.1% overall, $61.7M net loss [src-20260224-a7b149]. - Metal distributors: Reliance 27.9% or 31.6% (Q3 2026) [src-20261021-16bb64]; Ryerson 17.7% [src-20260729-407fd7]. Nucor 20%, including a one-off refund worth about 1.3 points [src-20260812-77d6f5]. - Howmet engine unit EBITDA margin 37.7% [src-20260806-9a83f7]; Carpenter 32.1% (40.4% ex-surcharges) [src-20261022-d47483]; Rockwell 49.5% [src-20260804-0e8130].

What expert calls add (claims, med). PCC wants accounts of $50M+; its share of Boeing machining fell from 50% to 30% in 10 years, and it is also automating [src-20250712-e5e110]. Boeing uses more than 10 machining shops and wants 2-3; old suppliers have “extremely high inertia” [src-20260818-9b2d5c]. How much Boeing makes itself is disputed (about 40% vs 20-30%) [src-20250712-e5e110] [src-20260818-9b2d5c].

Incumbents partner rather than build. Lockheed hosts Hadrian [src-20261014-015327]; RTX Ventures led Radical [src-20250718-d46f24] and backs Fabri [src-20260617-4dfb96]; Siemens bought Altair and Dotmatics [src-20250326-cc1810][src-20250701-a7fef9].

Failures to learn from. Markforged: $2.1B in 2021, sold for $42.5M in 2026; Velo3D’s shareholders were wiped out but the business recovered on defense [src-20261023-63156a][src-20261023-9f720f]. Bright Machines’ $1.6B SPAC fell through [src-20240527-6a4884]. Daedalus, an AI-run CNC factory in Germany, went insolvent in August 2026, the closest failed copy of the thesis [src-20260801-4532f6]. VulcanForms replaced its CEO, then raised $220M (a pivot, not a failure) [src-20240919-72efcc][src-20260130-365541].

What this means [inference]. The real threat is approved suppliers like PCC automating while keeping their approvals, and primes keeping critical work in-house.

8.1 The right to win: versus job shops, contract manufacturers and overseas

For. - **Old shops run like consultancies. Claim (Power):** reselling $60-an-hour labor at $120 [src-20250509-22b78a]; greenfield beat retrofit because upgraded certified shops were maybe 30% more efficient [src-20260908-65fab6]. - **One roof, good paperwork, responsiveness.** FlyX's CEO came back to Hadrian for "everything under one roof" and "proper documentation" [src-20260806-3a2ec7]; a Raytheon engineer saw lower part-to-part variation [src-20260823-b3713f]; an Anduril engineer praises 24-hour responses versus mom-and-pop "radio silence" [src-20250623-5bc67c]. - **Protected and urgent work stays onshore** (section 2.6). **Claim (Isembard):** lead time cut from about 10 weeks to under 14 days at matched price for one customer [src-20260310-cdfd82].

Against (equal weight). - **Machining is crowded.** "No shortage" of machine shops (Rolls-Royce engineer) [src-20251120-0c271b]; legacy Bender CCP holds $40M of a $944M Navy program [src-20261018-2b4a22]. - **Incumbents automate too.** Northrop claims about 20% lower recurring cost at a new plant [src-20261022-1389b7]; L3Harris credits AI for yields, deliveries up 60% [src-20261022-869770]. - **Buyers design for many vendors** and will not switch for "5%-10% cash back" [src-20250623-5bc67c]. - **China still wins on cost in open markets. Claim (Power):** US parts cost "two to three times" China's [src-20250509-22b78a]; China can deliver a forging "for the same cost that a U.S. forge will pay for the raw materials" [src-20220801-c32e19]. - **AI-washing.** Freeform's CEO: "We can slap iPads on things and epoxy floors"; "ask them exactly how" [src-20261020-bac9ca].

9. Customers and repeat orders

Who buys. Primes (Lockheed, RTX, Northrop, Boeing, L3Harris, General Atomics), new primes (Anduril), depots and programs (Army, Navy, DIU), and commercial hardware firms (SendCutSend serves 60% of the Fortune 500) [src-20261020-487a39].

What buyers told expert calls (med). Price evidence is in the decision summary; the rest: - Willingness to pay for speed: a Northrop engineer would pay 20-40% more to avoid a delay [src-20260802-36d46c]; ex-customer FlyX says "10%-15% is tops" and left over cost when work became "less stringent" [src-20260806-3a2ec7]. - Share today: Hadrian holds 2-3% of one Boeing unit's precision spend and under 10-20% of a Northrop unit's outsourced work [src-20260802-ef6506][src-20260802-36d46c]. Big buyers move work in-house after a "honeymoon period" [src-20250729-a0a78c]. - Raytheon reconciles "cheaper" with Boeing's "dearer at volume" if Hadrian prices low on small urgent jobs [inference].

Qualification: a tax at entry, a moat once paid. 9-12 months at Raytheon [src-20250708-2aa3ad], 1-2 years at Northrop [src-20260802-36d46c]; **Claim (Power):** full aerospace trust takes 7-8 years [src-20260908-65fab6]. Amca buys qualification by acquiring shops [src-20260520-af3e6c][inference].

Repeat orders in federal data. Firestorm alone climbed the full ladder (SBIR, then Phase III IDIQ, then 19 task orders, $53.8M) [src-20251022-629fc0][src-20251219-bd5788]. Hadrian's Army award ($39.2M obligated of $80M, one bid) is recorded as "DATA RIGHTS AND INTELLECTUAL PROPERTY", not plainly parts [src-20260317-5585be][src-20260317-77e798]. Mariana, Arda, 1872 and Nox: no prime awards found.

9.1 How buyers qualify suppliers and reorder, step by step

- **The ladder.** "A PO for six months … If they can do it well, giving another PO for two years"; prototypes are bought on lead time, production on price [src-20250623-5bc67c]. Any change of

design, material, process or site repeats first-article inspection [src-20190801-e81d7b].

- **Probation is small.** Raytheon spend on Hadrian is “in the tens of thousands … on purpose”, with “tens of millions” possible later [src-20260823-b3713f].
- **Buyers cap each supplier** at about 25-30% and want about 25% spare capacity [src-20250623-5bc67c].
- **“Field of dreams.”** The Pentagon “cannot actually sign a contract until they have that physical facility built” [src-20260325-7ba845]; Lockheed: the government does not “want to buy factories. We want to buy weapons” [src-20261014-47f64b].
- **Base rates.** About 5% of companies go from SBIR Phase II to Phase III [src-20240129-b46554]; Castelion calls the roughly 3-year gap the “valley of death” [src-20251010-d50ea5]. Ursa Major’s $137M and Divergent’s ~$3.64M Air Force awards are development-scale [src-20261014-08000a] [src-20250924-22c5ee].
- **What this means [inference].** Most new producers are on the lower rungs: probation and small repeat POs.

10. Capital and financing

How it works. Equity pays for software, team and plant #1; loans, government money and project finance pay for the rest, but only once contracts exist [inference]. **Claim (Power):** “equity never funds factory construction” [src-20260921-876080].

Company	Equity	Debt / government	Plant cost
Hadrian	$1.37B Series D at $7.87B [src-20260807-b0dc5b]	$360M credit line [src-20260817-c48f28]; $900M Navy [src-20260320-67ba4c]	~$200M for 270,000 sq ft [src-20250820-cfbf02]
Divergent	$290M Series E at $2.3B [src-20250915-311606]	$40M debt in that round	Undisclosed [src-20261023-e02543]
SendCutSend	$116M plus Oct round at $2.1B [src-20261020-487a39]	Equipment loans, ~$50M per plant [src-20260519-2d6ceb]	~$50M
Isembard	$59M [src-20260310-45bab3]	Undisclosed; franchisees fund their own [src-20250929-857f17]	n/a
Vulcan	$65M Series A + $50M Commerce equity	$620M OSC loan, $550M private [src-20251103-ee9b27][src-20261010-8e6ec0]	10,000-ton plant
Mariana	~$400M incl. project money [src-20260803-30e6d5]	Project finance	Per site

Bear. Hardware companies at Series B often have about $10M revenue and break-even gross margins [src-20251211-87dc10]. Asset-based lending is not evidenced for any tracked startup (gap).

11. Policy tracker (newest first)

Section 2.3 explains each tool; this is the dated list.

Date	Policy	What it does	Source
2026-11-27 (deadline)	China Ga/Ge/Sb export-ban suspension ends	Ban returns unless extended	[src-20261027-f82d55]
2026-11-10 (deadline)	China rare-earth controls suspension ends	Broad controls return unless extended	[src-20261019-854e22]

Date	Policy	What it does	Source
2026-11-01 (deadline)	OSC fund-finance applications due	$500M-$1B of loans per fund	[src-20260831-4dd3b7]
2026-10-20	Army plan for 1M drones (reported)	Volume demand for magnets and airframes	[src-20261020-878aa9]
2026-10-19	DPA Title III tally	23 awards, $820.5M since Jan 2025; none to tracked startups	[src-20261019-68fd3d] (med)
2026-10-08	Genesis Mission Phase II	12 awards, $159M	[src-20261008-452806]
2026-09-23	Army drone magnets	Vulcan chosen	[src-20260923-c53c99]
2026-09-16	FOMC hike	3.75-4.00%	[src-20260916-f0c016]
2026-06-18	OSC rare-earth loans	$725M Energy Fuels, $500M Phoenix Tailings (conditional)	[src-20260618-184e0c]
2026-04-06	Section 232 metals tariffs	50% / 25% / 10% tiers	[src-20260402-1b5d6c]
2026-01	IRS Notice 2026-11	100% bonus depreciation permanent	[src-20260114-2075f7]
2025-12-18	FY26 NDAA	Multiyear munitions (Sec. 804(b)); Industrial Base Fund purchase commitments up to 10 years with advance payments (Sec. 867); spare-parts review (Sec. 1842)	[src-20251218-f0d6ef][src-20251218-6c4228]
2025-11-03	OSC loan to Vulcan	$620M; the only federal loan to a tracked name	[src-20251103-ee9b27]
2025-08-01	Section 232 copper	50% on semi-finished copper	[src-20250805-80f2f4]
2024-12-19	OTA authority	Prototypes can roll into production without new competition	[src-20241219-9f8fcc]

Gaps. Golden Dome money to tracked names is unknown [src-20251218-f0d6ef].

12. Companies worth attention

“Target” = diligence call now; “Candidate” = watch; “Benchmark” = too expensive for SPC but useful for comparison. Profiles are in each chapter’s section x.10.

- **Targets:** Senra (founder call; first-pass yield and repeat orders), Freeform (customer references), Isembard (economics per franchise; how primes audit franchised sites; Thales is skeptical [src-20251110-84b67e]).
- **Enabler worth a call:** Lumaril, because it is paid on the production it adds (company claim) [src-20261027-d5ba3f]. Next: one plant’s before/after yield (7.10).
- **Candidates:** Earth AI (drill results, resource estimate), Fabri (a second source), 1872 (first repeat order), Atomic (a buyer call on repeat programs), Machina (verify valuation).
- **Benchmarks:** Hadrian, Divergent, SendCutSend, Mariana, Amca ($1B-$7.9B). Periodic, CuspAI, Mind and Arda are priced before product.
- **White space:** software for mid-size metal distributors (about 9,400 sites) [src-20251201-4b732d]; AI-native forging [src-20220215-45c353]; inspection as a service [src-20260908-ed94f0].

13. Bottom line

Five conclusions. 1. The category is real, but proven demand is narrow: speed on prototypes and low-rate work, at 2-10% of spend at the primes we spoke to [src-20260802-ef6506][src-20260802-36d46c]. 2. The data loop has partial evidence (score 2) only where the process is continuous (Freeform, Mariana, Radical). Nobody shows improvement over time. 3. Capital falls into three tiers: capital-light owner-operators (SendCutSend, Senra), debt- and government-funded heavy builders (Hadrian, Vulcan, Mariana), and software vendors that need someone else's plant (Arda, Mind) [inference]. 4. Precision & Defense Hardware is priced out at the top (Hadrian $7.87B [src-20260807-b0dc5b], Amca over $1B [src-20260520-af3e6c]); SPC's entry is Series B in fabrication or a hybrid/franchise model [inference]. 5. Policy money lands mostly on metals supply (OSC, tariffs, DPA) [src-20251103-ee9b27][src-20261019-68fd3d]; precision parts get their pull through prime contracts [inference].

Picks [inference]. Fabrication: Senra. Precision: Freeform; Isembard (hybrid). Metals: Vulcan and Mariana as benchmarks. Materials R&D: Radical and Periodic. Factory software: Arda and Mind as benchmarks.

Bias check. Senra (10 sources), Freeform (14) and Isembard (13) are among the least-covered names (Hadrian 76). Each pick rests mostly on company claims; treat all three as diligence hypotheses.

What would change our mind. A prime shows a new producer at over 20% of rate-production spend at a lower unit price (bull); a company publishes scrap, cycle time or quote accuracy improving with volume (bull); Section 232 tariffs are rolled back (bear); a tracked owner-operator raises down or stalls a plant (bear).

13.1 The bear case in one list

Points already made above are listed by section. - Buyers pay parity or more, and speed matters to few (decision summary 1). - The model has failed recently: Daedalus, Bright Machines, Markforged (section 8). - Quality on complex parts is unproven and regulators tighten after Boeing's failures: "Innovation … is the first thing that gets sacrificed" [src-20260818-9b2d5c]. - The AI is thinner than the pitch: Opus relies on outside CAM [src-20261004-9b47f5]; robot learning succeeds about 60% of the time versus 99.9% needed [src-20260904-267293]; A-Lab's 41 new compounds were challenged [src-20240124-513fd9]; CloudNC sells AI CAM to 1,000+ shops including Lockheed's [src-20260908-f98d21]. - Valuations run ahead of audited revenue (Hadrian $7.87B, Mind $3.4B, Periodic about $7.5B reported); plants built ahead of contracts are exposed to a defense or tariff reversal [src-20260921-876080]. - Spare capacity exists: existing spindles cut metal only 34% of the time [src-20250120-419768]. - Share per customer is capped and repeat orders are tiny (sections 9 and 9.1). - Defense demand is lumpy and budget-gated: Carpenter's defense bookings are down because funds haven't reached customers [src-20261022-de5e6c]; 82% of the FY27 munitions request rests on reconciliation [src-20261007-c6f64e]. - Incumbents are booming: Carpenter's specialty-alloy margin hit 37.7% [src-20261022-679fcc]; Reliance is taking share with no AI [src-20261022-1e6872]. - Teams churn: 90% of Hadrian's first 10 employees left by year 5 [src-20260817-8867f2]. Some see a debt business, not a venture one [src-20250709-43486e]. - Commodity risk: Ganfeng's market value fell more than 75% with lithium prices [src-20250529-3a588b].

Appendix A. Glossary

Term	Definition
A&D	Aerospace and defense
ABL	Asset-based lending: credit secured by receivables and inventory

Term	Definition
ACV	Annual contract value
AD/CVD	Antidumping and countervailing duties on unfairly priced imports
AM	Additive manufacturing: 3D printing
AS9100	Aerospace quality management standard; table stakes for flight parts
AS9102	Standard form for a first article inspection report
B2B / B2G	Selling to businesses / to government
BOM	Bill of materials
CAM	Software turning a 3D model into machine toolpaths
Capex	Capital expenditure: money spent on plants and machines
CHIPS Act	2022 law funding US chip factories ($52.7B)
CMMC	DoD cybersecurity certification for contractors
CNC	Computer numerical control: a computer-driven cutting machine
CRO	Contract research organization: an outside lab hired for experiments
CSO	Commercial Solutions Opening, a fast DoD solicitation
DFARS	DoD procurement rules, incl. specialty-metal sourcing
DIB	Defense industrial base: every company that supplies the Defense Department
DIU	Defense Innovation Unit, which buys commercial tech fast
DoD / DoW	Department of Defense, also called Department of War since 2025
DPA Title III	Defense Production Act authority to fund domestic capacity
EBITDA	Earnings before interest, tax, depreciation and amortization; a rough cash-profit measure
FAI	First article inspection: full check of the first part made
FFP	Firm fixed price contract
Form D	SEC filing a company makes when it sells private shares
FPY	First-pass yield: share passing inspection first time
GM	Gross margin: revenue minus direct cost, as a share of revenue
IDIQ	Indefinite-delivery, indefinite-quantity contract vehicle
IEA	International Energy Agency
IRA / 45X	Inflation Reduction Act (2022); 45X is its per-unit manufacturing tax credit
ITAR	US export control limiting defense work to US persons
Job shop	Small owner-run shop building to a customer’s drawing
LIFO/FIFO	Inventory accounting methods; LIFO lowers margins when prices rise
LPBF	Laser powder bed fusion, a metal 3D printing process
LRIP	Low-rate initial production
MES	Software that tracks and directs shop-floor work
NADCAP	Accreditation for special processes (heat treat, NDT, coatings)

Term	Definition
NDAA	National Defense Authorization Act, the yearly defense policy law
NDT	Non-destructive testing
Nesting	Packing parts on sheet or stock to minimize scrap
NRE	Non-recurring engineering: one-time setup cost before production
ODM	Original design manufacturer: builds products others sell under their brand
OSC	DoD Office of Strategic Capital, a lender
OTA	Other Transaction Agreement, fast non-FAR prototype contract
PPBE	DoD planning, programming, budgeting and execution process
Prime	Top contractor selling whole systems to the government (Lockheed, RTX, Boeing)
RFQ	Request for quote
RL	Reinforcement learning: AI that learns by trial and reward
SBIR	Staged federal R&D grants to small businesses
Section 232	National-security tariff authority (1962 statute)
Service center	Metal distributor that stocks, cuts and ships
SPAC	A shell company that lists a private company on the stock market
TAM	Total addressable market
Tier-1/2/3	Supplier levels below a prime: sub-systems, then parts, then raw inputs
USAspending	Public federal database of contract awards
Utilization	Share of available machine hours actually producing

Appendix B. Gaps and next actions

- Hadrian revenue: two low-confidence figures conflict; get a primary number.
- Startup gross margins: none disclosed; ask Senra, SendCutSend.
- Raw Materials (Stage B): refining TAM and economics: no free source found.
- Manufacturing: fabrication TAM: none; Census CBP gives counts and payroll only [src-20250626-23f6d9]. Next: 2022 Economic Census receipts by NAICS 3323/3327.
- AI/robotics cost decline over 24 months: no source.
- Golden Dome amounts: the NDAA funding tables reference Golden Dome funds; size and recipients unconfirmed [src-20251218-f0d6ef]. Freeform Skyfall go-live: unconfirmed.
- ABL use by startups: none evidenced.
- Exits (Q14): no comparable acquisition above $100M in repo.
- Specialty Chemicals: global market estimates range from $626.5B to $1.4T depending on scope (5.2); no clean US figure.
- Raw Materials (Stage A): Earth AI has no published resource estimate; its 75% success rate is self-reported and undefined [src-20250127-c28430].
- Advanced Materials: no market size for materials R&D or US cathode demand in dollars.
- Enabling Software: Lumaril customers, revenue and measured yield gains are not public; WSJ article body is paywalled (headline only) [src-20261027-79ddd6].

Appendix C. Source coverage

Per-source ledger: `insights/exports/primer-ledger.csv`. All 28 on-thesis expert calls and all 102 podcasts were read in full; the ~200 cited sources were read in full except long filings (FY26 NDAA, DIU guidebook, boilerplate in 10-Q/10-K filings), which were checked against each claim. Podcast claims are speakers' words on friendly, sometimes sponsored or investor-hosted shows, and are labeled as claims.

Appendix D. Listening list

Every link, with length and source ID, is in `insights/listening-list-by-industry.md` (exported as VII-listening-list-v2). Each chapter's "Listen & watch" section lists that chapter's episodes; transcript summaries are in VII-podcast-summaries.